Funding Instruments and Effort Choices in Higher Education *

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Abstract

This paper examines the effects of Tuition-Free Higher Education (TFHE) policies on student enrollment, persistence, and academic performance, with a focus on Chile's 2016 *Gratuidad* reform, which granted free higher education to students from the lowest five income deciles. Using a difference-in-differences design, we find that the policy raised enrollment among eligible students by around 10 percentage points, with the response concentrated among lower-ability students previously excluded from public funding. Among students whose enrollment decision was already made before the reform, dropout fell by up to 2.5 percentage points and on-time graduation rose by 1.8 percentage points. For post-reform cohorts, aggregate completion rates are essentially unchanged despite a leftward shift in the ability distribution of entrants. To disentangle selection from behavioral responses, we develop and estimate a dynamic discrete-choice model of enrollment, program choice, and academic effort. We find no evidence that the removal of performance requirements weakened effort among infra-marginal students. Instead, the composition of new entrants induced by the reform, who exhibit higher fixed and marginal costs, accounts for most of the post-reform progression patterns. TFHE thus expanded access without compromising aggregate educational quality.

Keywords: Financial aid, Effort choices, Higher Education
JEL Codes: I22, I23, I26

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1 Introduction

Higher education is a central determinant of long-run earnings, inequality, and intergenerational mobility. Despite decades of expansion in attainment, the returns to a college degree have remained large (Card, 1999; Goldin & Katz, 2008), and access to selective institutions is a powerful lever for upward mobility (Chetty et al., 2020). Yet completion is unequally distributed, and the gap in degree attainment between high and low-income youth has widened in recent decades (Bailey and Dynarski, 2011, S. Dynarski et al., 2022). In response, governments have invested heavily in policies aimed at lowering financial barriers to college. As of today, more than 40 countries operate some form of tuition-free higher education (TFHE) policy, generally limited to public institutions, and recent debates in the United States over student-debt forgiveness, free community college, and need-based grants reflect a similar emphasis. Public spending on higher education in the OECD has risen accordingly (OECD, 2019), raising the question of how aid should be designed to balance equity, fiscal cost, and educational outcomes.

Higher education policies operate on two margins. On the extensive margin, lowering net tuition draws in marginal students who would otherwise have stayed out. On the intensive margin, the design of aid shapes effort, persistence, and completion among those who do enroll. Traditional instruments combine financial relief with performance requirements. The Pell Grant in the United States is conditioned on satisfactory academic progress, and merit-based scholarships typically require a minimum GPA to renew (Scott-Clayton, 2011, Deming and Dynarski, 2010). Subsidized loans expand access but burden graduates with debt that distorts post-college choices (Rothstein and Rouse, 2011, Sieg and Wang, 2018, Black et al., 2023). TFHE sits at one extreme of this design space, eliminating out-of-pocket tuition for a broad eligible population without imposing progress conditions, and thus removing both a financial barrier and a performance lever simultaneously. The ex-ante effect on educational outcomes is therefore ambiguous. Eliminating tuition relaxes a binding budget constraint, frees up time previously devoted to part-time work, and reduces dropout after negative income shocks. At the same time, moving from a conditional to an unconditional scheme may weaken incentives for effort among inframarginal students (Scott-Clayton, 2011), while drawing into the system a less-prepared pool of marginal entrants. Consistent with this tension, countries with TFHE tend to exhibit lower aggregate graduation rates than comparable countries in which students face out-of-pocket fees, but aggregate measures conflate the two responses and cannot answer whether TFHE produces more or fewer graduates per dollar spent.

This paper studies the effects of a large-scale TFHE reform on enrollment, persistence, and academic performance, and develops a framework that disentangles selection from behavioral responses. We proceed in two steps. First, we exploit Chile's 2016 *Gratuidad* reform, which granted tuition-free higher education to students from the lowest five income deciles in roughly half of the operating universities and vocational centers (including some private institutions), as a natural experiment, and use a difference-in-differences design to estimate effects on enrollment, persistence, dropout, and graduation. Second, we build and estimate a dynamic structural model of enrollment, program choice, and academic effort that endogenizes how students respond to changes in out-of-pocket tuition and to the presence or absence of performance requirements. Together, the reduced-form and structural exercises decompose aggregate impacts into a composition effect from marginal students induced to enroll and a behavioral effect on the effort choices of infra-marginal students.

The Chilean context offers an ideal setting for this analysis, thanks to both rich administrative data and a well-defined policy intervention. We assemble administrative records that link the universe of national-exam takers to higher education enrollment, scholarship assignment, course-by-course credit completion, dropout, and graduation outcomes. The *Gratuidad* reform is a sharp, large-scale policy change with a clearly defined eligibility rule based on income decile, introduced on top of a well-developed pre-existing aid system that included subsidized loans (the State Guaranteed Loan), need-based scholarships, and a centralized admission platform. Crucially, the pre-existing instruments were all performance-conditional. Subsidized loans required scoring above 475 on the national entrance exam (PSU), and the main scholarships required a PSU score above 510 and the completion of at least 70% of registered credits each year. *Gratuidad* replaced these conditional schemes with full, unconditional tuition coverage for eligible students. The reform therefore varied both the level of net tuition and the performance conditioning of aid, providing the variation needed to identify selection and effort responses separately.

The difference-in-differences estimates reveal a sizable extensive-margin response. Enrollment among eligible students rose by 3.5 percentage points (p.p.) in 2016 and stabilized at around 10 p.p. from 2017 onward, when the policy was extended to vocational centers. The increase is concentrated among lower-ability students who, prior to the reform, did not qualify for either subsidized loans or merit scholarships. Their enrollment rose by 14 p.p., compared with 5.5 p.p. for students already eligible for the most generous pre-existing aid. We find no evidence of crowding-out at non-eligible institutions, in contrast to the spillover patterns documented by Londoño-Vélez et al. (2020) in Colombia, suggesting that the bulk of the response comes from genuinely marginal students rather than substitution across institutions. As a result, the composition of incoming cohorts shifted to the left of the ability distribution, a pattern with first-order implications for downstream outcomes.

Estimating effects on educational outcomes is more delicate, because the change in composition confounds the response of infra-marginal students with that of new entrants. We exploit a feature of the policy design to address this. Cohorts that had already enrolled before 2016 became eligible for *Gratuidad* on their remaining tuition payments, with treatment intensity varying by cohort but with enrollment decisions already made. This holds composition fixed and isolates the response of infra-marginal students. Among these students, the policy reduced dropout by up to 2.5 p.p. and increased on-time graduation by 1.8 p.p. For 2016-and-after cohorts, by contrast, we find essentially null effects on dropout, graduation, and persistence, despite the negative composition shift. Two forces operate in opposite directions. Infra-marginal students benefit from lower tuition and from the removal of the threat of losing the scholarship, while new marginal entrants progress more slowly. The reduced-form evidence is consistent with the two effects approximately offsetting one another, leaving aggregate completion rates roughly unchanged but the absolute number of graduates higher.

While the reduced-form analysis shows that the policy was successful in expanding access without lowering aggregate completion rates, two reasons motivate a structural approach. First, the difference-in-differences estimates cannot decompose, for each aggregate outcome, the relative contribution of marginal students induced to enroll and infra-marginal students who change their behavior. Second, the policy bundled two changes that the data cannot separately identify without further structure, namely the reduction in out-of-pocket tuition and the elimination of performance requirements. Recovering the underlying cost primitives is what allows us to attribute observed changes in progression to selection on costs and ability or to changes in effort, and to interpret post-reform behavior in a setting where the price of tuition and the structure of incentives moved jointly.

For these reasons, we develop a dynamic discrete-choice model of enrollment, field and institution choice, dropout, and academic effort. In the first stage, a student observes her type (gender, ability, and socioeconomic status) and her personalized choice set (Fack et al., 2019), and decides whether and where to enroll. Conditional on enrollment, in each subsequent period she chooses whether to continue and how much effort to exert, where effort shapes the distribution of credit completion. Performance feeds back into next-period out-of-pocket tuition through the rules of pre-existing scholarship programs, generating a dynamic link between current effort and future financial incentives. Effort is unobserved and recovered from a first-order condition following de Groote (2025), which allows the marginal cost of effort to be estimated jointly with the fixed costs of enrollment. Identification of effort responses relies on the quasi-experimental variation in out-of-pocket tuition induced by the differential timing of the reform across cohorts, together with the performance-conditional nature of the pre-existing aid system.

Estimation reveals substantial heterogeneity in costs across the student distribution. Fixed costs increase with out-of-pocket tuition, especially at the enrollment stage. They are also higher for female students, for students enrolling in a different region, and for students who have fallen behind in their program. Marginal costs of effort exhibit a clear gradient in both ability and income, with lower-PSU and lower-income students facing higher marginal costs of producing a unit of academic progress. Critically, we find no evidence that the removal of performance requirements weakened effort among inframarginal students. The composition of new entrants induced by *Gratuidad*, who exhibit higher fixed and marginal costs, accounts for most of the post-reform progression patterns, while the behavior of incumbent students changes little. The model therefore rationalizes the reduced-form evidence: aggregate completion is held in check by composition rather than by moral hazard.

Related literature This paper builds upon several strands of literature. It first contributes to the literature on higher education financing and student outcomes. Broad reviews emphasize that aid design affects not only access, but also persistence, completion, and the timing at which students learn about prices and eligibility (Deming and Dynarski, 2010, Page and Scott-Clayton, 2016, S. Dynarski et al., 2022). There is substantial evidence showing that increasing grants or reducing net tuition raises enrollment and attainment (S. M. Dynarski, 2003, Castleman and Long, 2016, Angrist et al., 2015, Denning, 2017, Fack and Grenet, 2015, Londoño-Vélez et al., 2020, Dobbin et al., 2022, Harris and Mills, 2025). Comparative evidence from Solis, 2017, Molina and Rivadeneyra, 2021, Duryea et al., 2023, and Bolhaar et al., 2024 further shows that the effect of subsidies depends on the pre-reform financing environment and on whether aid takes the form of grants or loans. Evidence on completion and performance is more mixed once composition effects are taken into account (Cohodes and Goodman, 2014, Denning, 2019, Denning et al., 2019, M. M. Ferreyra et al., 2024). Beyond partial equilibrium responses, Epple et al., 2006, Epple et al., 2017, and Abbott et al., 2019 show that equilibrium supply responses and redistribution matter for the incidence of education-finance reforms. This paper complements that literature by studying a large-scale tuition-free reform in a setting where the public aid system was already well developed, and by tracing both extensive-margin enrollment responses and post-enrollment outcomes.

It also contributes to the growing literature on tuition-free college and the design of free-college programs. Bucarey (2017) studies the crowding-out effects of Chile's tuition-free reform on low-income students at selective programs. Recent work emphasizes that free-college design is consequential. Unconditional or early-committed aid can generate larger take-up responses than aid that remains uncertain or application-intensive (Burland, 2023, Harris and Mills, 2025), while dynamic models imply that free college often raises enrollment more than graduation because outcomes depend on the type of students induced to enroll and on their effort choices (Matsuda, 2020, M. M. Ferreyra et al., 2024). Comparative evidence from promise programs and tuition reversals also documents meaningful heterogeneity across institutional settings (Carruthers and Fox, 2016, Attridge et al., 2025, Bietenbeck et al., 2023). Our contribution is to bring this design question to the Chilean reform and to quantify how much of the observed effect operates through selection versus changes in effort among infra-marginal students.

Second, we contribute to the literature on the incentive structures and moral hazard inherent in educational funding. Traditional subsidies lower the cost of entry, but they may also weaken the marginal cost of delay or failure. Scott-Clayton (2011) shows that academic incentives embedded in aid rules affect college achievement. Garibaldi et al. (2012) and Beneito et al. (2018) show that continuation tuition and retake pricing can significantly reduce delay and increase effort, while Montalbán (2022) finds that cash support is most effective when accompanied by demanding progress requirements. Related evidence on aid packaging and borrowing frictions (Marx and Turner, 2018, Falco and Reichlin, 2025) suggests that students respond both to the level and to the composition of aid. TFHE also releases students from debt accumulation, which has been shown to affect major choice (Rothstein and Rouse, 2011), dropout (Stinebrickner and Stinebrickner, 2008), career decisions (Sieg and Wang, 2018), and post-college household formation (Black et al., 2023). This paper speaks directly to that mechanism because the Chilean reform simultaneously changed out-of-pocket tuition and removed academic-progress requirements for a large group of students.

Finally, we contribute to a growing structural and semi-structural literature on educational choices under dynamic incentives. A number of papers study responses to changes in incentives by modelling effort or progression as part of a dynamic problem, including

Arcidiacono (2005) for affirmative action and admissions, Beffy et al. (2012) for study duration under changing returns to education, and Joensen and Mattana (2024) for grants, loans, and academic progression. Closest in spirit are M. Ferreyra et al. (2022) in Colombia and Tincani et al. (2023) in Chile, both of which allow incentives and preparedness to shape progression. In a related Chilean setting, Larroucau and Rios (2024) studies retention and reapplication decisions in a centralized admissions market. In our paper, effort is endogenous and recovered from a first-order condition, following de Groote (2025). We contribute by linking current effort to future educational costs through out-of-pocket fees and performance requirements, which allows the model to decompose adverse selection from moral hazard under a large-scale tuition-free reform.

Overview The rest of the paper is organized as follows. Section 2 describes the institutional details, the TFHE policy and introduces the data. Section 3 reports results from differences-in-differences estimation, and Section 4 builds a model of enrollment and effort in higher education. Section 5 discusses the main results of the model. Section 6 provides a discussion and next steps.

2 Setting

This section describes the institutional setting, the policy of interest, and the data used in our analysis. Section 2.1 provides an overview of the Chilean higher education system and the pre-existing funding instruments. Section 2.2 describes the Tuition-Free Higher Education policy introduced in 2016 and the substitution patterns it generated across the previously available aid programs. Section 2.3 introduces the administrative records that we use to track enrollment, performance, and graduation outcomes, and defines the samples used in the reduced-form and structural analyses.

2.1 Institutional Setting

Chilean higher education is well developed, with enrollment rising steadily since the General Law of Universities (*Ley General de Universidades*) of 1981, which incentivized the creation of institutions by allowing entry without state dependence. In 2023, the system comprised a total of 138 institutes: 80 vocational schools (33 Professional Institutes and 47 Centers of Technical Formation) and 58 universities.[1]

[1]https://www.ayudamineduc.cl/ficha/instituciones-vigentes-reconocidas-por-el-mineduc.

Universities are divided between the so-called *traditional* universities (a mix of public and private institutions that receive direct support from the government) and *private* universities, which constitute the rest. The latter were created after 1981 and are mainly financed through tuition. Traditional universities are officially known as the Universities of the Rector's Council (*Consejo de Rectores de las Universidades Chilenas, CRUCH*) and are responsible for coordinating the higher education system. They comprise 30 universities and account for around half of total enrollment.[2]

Access to higher education is through a Centralized Admission Platform, compulsory for traditional universities and some private institutions. Vocational program admission is also possible, although most are conducted off-platform. After high-school completion, students applying to higher education take the centralized admission exam (*Prueba de Selección Universitaria,* PSU). This standardized test, similar to the SAT in the United States, ranges from 150 to 850 points, with a mean of 500 and a standard deviation of 110. Students then submit rank-ordered lists of up to 10 degree programs through the centralized system and are assigned via a deferred-acceptance algorithm (Gale & Shapley, 1962). In practice, the platform imposes a minimum PSU score of 450 to apply.

Higher education is costly in Chile. In 2009, the average tuition fee for a university program was equivalent to 47% of the median family income (Solis, 2017). Costs vary across institution types but remain high even in public institutions. Different funding instruments coexist. Students rely mainly on loans and grants from the Ministry of Education. Eligibility criteria are strongly related to PSU scores. The State Guaranteed Loan program (SGL), introduced in 2006, finances 90% of reference tuition and requires a PSU score of 475 or higher, with no socioeconomic requirement since 2014. Several scholarships also exist, with the *Beca Bicentenario* and *Beca Excelencia* being the most popular. On average, scholarships finance 80% of reference tuition. Eligibility requires a PSU score between 510 and 550, depending on the scholarship, and excludes students in the top two or three income deciles. Both short-cycle programs (SCPs) and university degrees are covered by scholarships. Two-year programs typically require a high-school GPA above 5.0. Access to private loans is limited: in 2015, only 7.5% of student loans came from banks without a state guarantee.

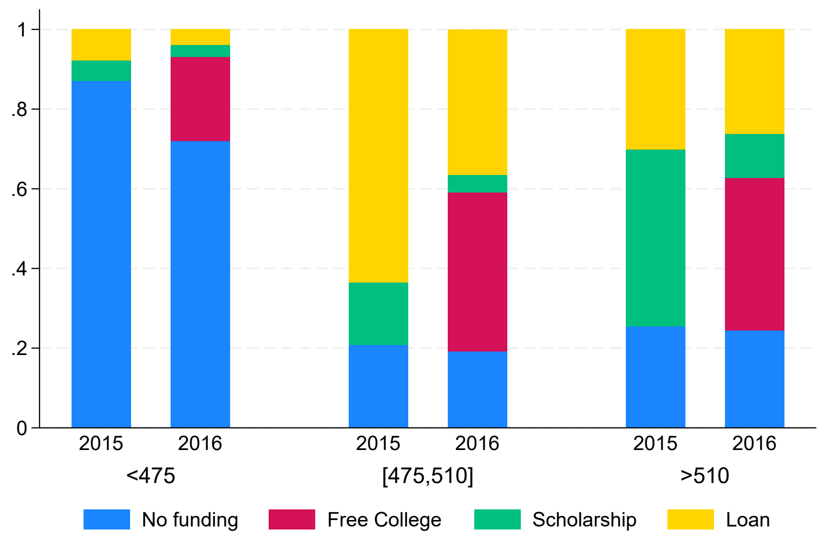
[2]In 2018, the government established a clear set of rules for being considered a traditional university with the promulgation of Law 21091, allowing the first three non-traditional universities to enter.

2.2 Tuition-Free Higher Education (TFHE) program

Since the implementation of SGL, student debt rose at a rate of 70% annually, and the number of recipients increased steadily from 15,800 in the first year to 652,000 students in 2016 (Bucarey et al., 2020). In 2011, students led mass protests demanding more affordable higher education. Michelle Bachelet was elected president in 2014 on a platform that included making college free by 2020. In 2015, the TFHE law was enacted, removing tuition fees for students in the bottom half of the income distribution. The policy was first implemented for the 2016 university cohort, expanded in 2017 to include vocational institutes, and further extended in 2018 to the sixth income decile. Applications for TFHE, loans, and scholarships are made through the Centralized Admission Platform by completing a short online form during the application period, regardless of whether the applicant applies to a platform degree or not. A schematic visualization of the policy is provided in Figure A4.

The introduction of TFHE policy generated different substitution patterns depending on students' test scores. Figure 1 shows the distribution of funding instruments before and after the policy, by test score segment. Students scoring below 475 were ineligible for subsidized loans or merit-based scholarships (except for a small scholarship targeting top students from low-SES schools). For this group, TFHE reduced the share of students with no funding. Students scoring between 475 and 510 qualified for subsidized loans but remained below the threshold for most scholarships; for them, substitution was mainly from loans to TFHE, relieving them of any debt upon graduation. Students scoring above 510 qualified for merit-based scholarships (conditional on income eligibility). For this group, substitution was mainly from scholarships to TFHE, which had two advantages: first, scholarships typically covered only 80% of tuition and fees, whereas TFHE provided full funding; second, scholarships required students to meet satisfactory progression standards (validating about 70% of annual course credits), while TFHE imposed no such requirements. Before the policy, about 20% of scholarship beneficiaries lost eligibility; of these, around 40% switched to subsidized loans, while the rest continued without public funding. After the policy, most substituted to TFHE, while those above the income threshold continued either with loans or no funding, in roughly the same proportions as before.

Figure 1: Distribution of funding instruments, by test score



Notes: This figure shows the distribution of funding instruments by test score group, for 2015 and 2016. PSU is the national standardized test. Students with test scores below 475 are not eligible for public funding, students scoring between 475 and 510 are eligible for loans (CAE and SGL), and students scoring above 510 are eligible forscholarships.

2.3 Data

We use Chilean administrative records that provide detailed student and program data. We observe enrollment, admission test scores, scholarship assignment, socioeconomic information, and demographic characteristics. Institutional data include type, tuition, location, and program length. Between 170,000 and 180,000 students take the national exam each year, of whom about 60% enroll in higher education. Descriptive statistics are reported in Table B1.

In addition, we use restricted data on credit completion from the Information Service of Higher Education (SIES) covering the universe of enrolled students from 2016 to 2023. These data report the number of courses registered and whether they were completed.

Quasi-Experimental sample We restrict the sample to first-time test takers for the 2012 to 2017 cohort. Since we observe the data until 2023 and the standard university program lasts five years, we allow seven years for students to complete their program. In 2012, scholarships were expanded to include the third income quintile, such that all eligible students for TFHE were already eligible for scholarships conditional on scoring a certain PSU score (Bucarey, 2017).

Model sample We are constrained by the fact that credits are only available from 2016. Together with allowing students 7 years for graduation, we estimate the model on a random sample of 2016-2017 cohorts. CCPs are estimated using the *universe* of 2016-2017 cohorts.

3 Policy Effects

This section provides reduced-form evidence on the effects of the TFHE policy on enrollment and educational outcomes. Section 3.1 presents the differences-in-differences specification and discusses the identifying assumption. In Section 3.2, we estimate the extensive-margin response and document substantial heterogeneity across ability groups and institution types. Section 3.3 addresses the composition problem that arises when estimating intensive-margin effects, and isolates the response of infra-marginal students by exploiting the differential exposure of pre-2016 cohorts to the policy. Section 3.4 discusses the main threats to identification and presents robustness checks.

3.1 Empirical Strategy

We are interested in causally estimating whether the TFHE policy had an impact on enrollment and educational outcomes. To do so, we exploit the exogenous variation in eligibility introduced by the policy and use differences-in-differences. We estimate the following regression model by OLS:

$$Y_{it} = \sum_{\substack{k=2011 \\ k \neq 2015}}^{2020} \beta_k \left(\mathbb{1}\{t = k\} \times \mathbb{1}\{dec_i < 6\} \right) + \gamma \, \mathbb{1}\{dec_i < 6\} + \delta_{\mathbf{t}} + \mathbf{X}_{\mathbf{i}}' \alpha_{\mathbf{x}} + \epsilon_{\mathbf{it}} \tag{1}$$

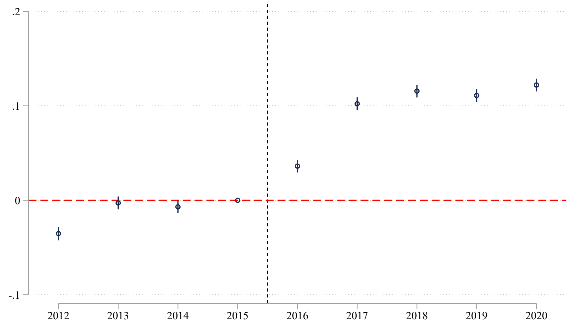
Y_{it} is an outcome of interest including enrollment and educational outcomes such as dropout, graduation, on-time graduation and persistence, defined as the time (in years) spent in college. $\mathbf{X}_{\mathrm{i}}'$ includes a constant, gender, PSU score, high school GPA, degree type, and mother education. $\mathbb{1}\{dec_i < 6\}$ determines eligibility to TFHE, defined as belonging to the lower half of the income distribution. δ_t are cohort fixed effects. We cluster standard errors at the income decile level using wild bootstrap (Cameron et al., 2008). We use the year before the implementation (2015) as the reference category. β_t is the differences-in-differences estimator.

The identifying assumption is that, in the absence of treatment, eligible and non-eligible students would have experienced similar trends in the outcomes of interest ($\beta_k < 2015$). We test for the validity of this assumption by examining pre-treatment trends. Additionally, we perform several robustness checks, including controlling for differential trends by income decile and restricting the sample to adjacent income deciles. The differences-in-differences estimator can be interpreted as the intent-to-treat effect (ITT) of the policy, given that not all eligible students actually receive TFHE (some do not enroll, while others enroll in non-eligible institutions).

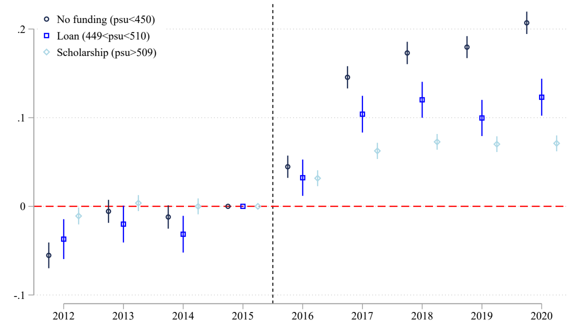
3.2 Effects on enrollment

We first examine the impact of TFHE on the extensive margin, i.e., enrollment. The 2016 cohort is the first for which the policy could affect the enrollment decision. Figure 2 shows the results of estimating Equation 1. As shown in panel a), enrollment among eligible students increased by 3.5 p.p. in 2016 and by around 10 p.p. from 2017 onwards, relative to 2015. This increase in enrollment comes from students for whom the returns to education became positive or who were previously credit constrained. The larger effect, starting in 2017, is consistent with the expansion of free higher education to SCPs. As explained in section 2, the control group are students in the upper half of the income distribution. We then examine heterogeneity by ability groups, which determine loan and scholarhip eligibility. In panel b), we observe that lower-ability students, who had limited access to public funding before 2016, are the group that increases enrollment the most. Differences between groups are significant in 2017, when lower-ability students increase enrollment by 14 p.p., compared to 5.5 p.p. for students who already had access to merit scholarships.

Figure 2: Enrollment



a) Effect on enrollment

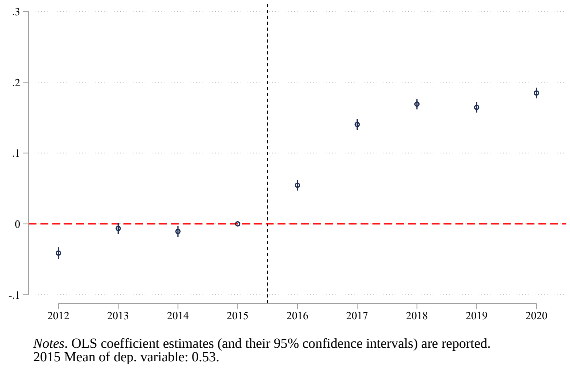


b) Effect on enrollment by funding eligibility

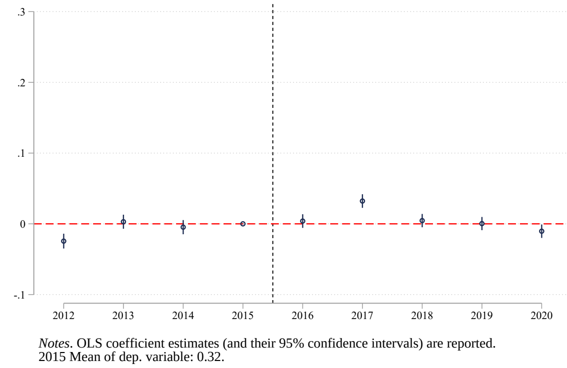
Notes: These figures show the results of estimating Equation 1. Sample is the universe of first time test takers. Panel (a) shows the aggregate results, while panel (b) estimates the equation for groups based on their test scores. PSU is the national standardized test. Students with test scores below 475 are not eligible for public funding, students scoring between 475 and 510 are eligible for loans, and students scoring above 510 are eligible forscholarships.

Substitution between eligible and non-eligible institutions Since TFHE affected around 50% of institutions and 70% of enrollment, it is also important to examine whether there was substitution from non-eligible to eligible centers, or whether enrollment is driven completely by marginal students. Figure 3 reports the estimation on enrollment for eligible and non-eligible institutions. Results show a significant increase of eligible students to higher educations institutes that were eligible to the policy. In panel a), we observe increases of 5 and 13 p.p. for years 2016 and 2017, respectively. In contrast to Londoño-Vélez et al. (2020), panel b) suggests no crowding-out. In fact, we find mostly a null effect on enrollment (except for 2017). When looking at panel c), we observe that the increase in enrollment in eligible institutions is driven by low-ability students. Panel d) decomposes the null effect on noneligible institutions in a decrease in enrollment coming from high-ability students while low-ability students increase enrollment around 2 to 4 p.p. A possible explanation is that the policy generated awareness about higher education opportunities and low-ability students that had enough resources opted for higher education even if not eligible to TFHE.

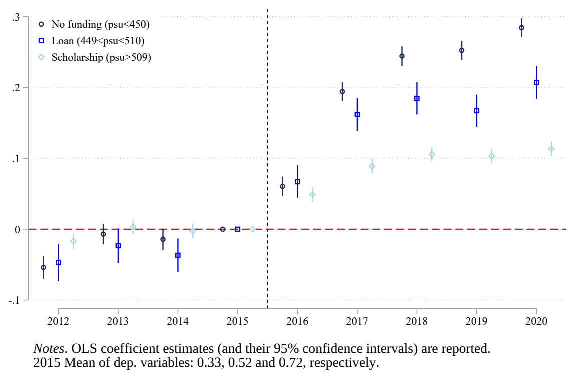
Figure 3: Enrollment by institution eligibility



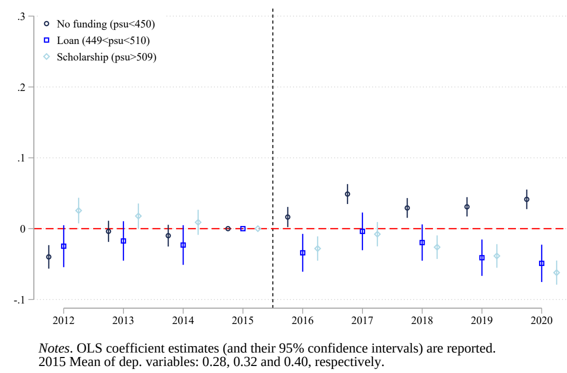
a) Eligible institutions



b) Non-eligible institutions



c) Eligible institutions by funding eligibility



d) Non-eligible institutions by funding eligibility

Notes: These figures show the results of estimating Equation 1 on enrollment for eligible and non-eligible institutions to the TFHE policy. Sample is the universe of first time test takers. Panel a) and b) show the aggregate results, while panels c) and d) disaggregate results by test score group.

Field Preferences We next look at whether students' preferences changed as a result of the policy. We use students' rank-ordered list and construct the dependent variable as the share of a field in student's list. We restrict the sample to eligible institutions and estimate Equation 1 for each field. Results are presented in Figure A5. Panel a) shows that TFHE increased preferences for Health by 1.5p.p. in 2016 and 2p.p. in 2017, while decreasing preferences Arts and Humanities by around 2p.p. STEM majors also lose popularity starting in 2018 with respect to 2015 by 2 p.p. Panel b) decomposes the results into finer categories. The decreases in Arts and Humanities are driven by Education, which decreases by 1p.p., 10% with respect to 2015 average. These results match the evidence found by Castro-Zarzur et al. (2022), which finds that TFHE eclipsed an incentive program to attract talent in Education majors [3]. The increase in Health preferences can be explained by the fact that they are the most expensive programs, along with law majors. Finally, the the 2p.p. decrease in STEM majors is driven by Technology.

3.3 Effects on educational outcomes

Recovering treatment effects for the intensive margin is not as straightforward. If we were to estimate educational outcomes using Equation 1, composition and moral hazard could difficult interpretation: the distribution of students' test scores shifted to the left starting 2016, given that a substantial fraction of lower-ability enrolled (marginal students) as can be seen in Figure A2. On top of that, 2016 cohort onwards eligible students were not subject to performance requirements anymore. To keep the composition of students fixed, we exploit a particularity of the policy: cohorts prior to 2016 were eligible to TFHE for the remaining years of the nominal length of their degrees. For example, if they enrolled in 2015 in a five-year degree, they can be exempted of tuition for the remaining four years, two years for the 2015 cohorts, etc.

We can use this exogenous variation to partial out selection. 2012 to 2015 cohorts are potentially eligible to TFHE starting 2016 but cannot change their initial enrollment decision by construction. We adapt Equation 1 as

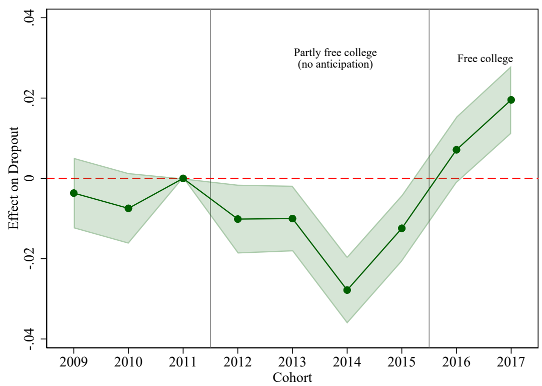
$$Y_{it} = \sum_{\substack{k=2010 \\ k\neq 2011}}^{2017} \beta_k^c \left(\mathbb{1}\{t=k\} \times \mathbb{1}\{dec_i < 6\}\right) + \gamma^c \, \mathbb{1}\{dec_i < 6\} + \delta_t^c + \mathbf{X_i}'\boldsymbol{\alpha}_x^c + \epsilon_{it}^c \tag{2}$$

The above regression remains a standard TWFE differences-in-differences regression, as the time dimension are cohorts. Treatment varies in intensity by cohort, but effects are always estimated with respect to a never treated cohort, so there are no issues of negative weighting (Callaway et al., 2024; de Chaisemartin & D'Haultfœuille, 2024). The variation exploited is purely cross-sectional.

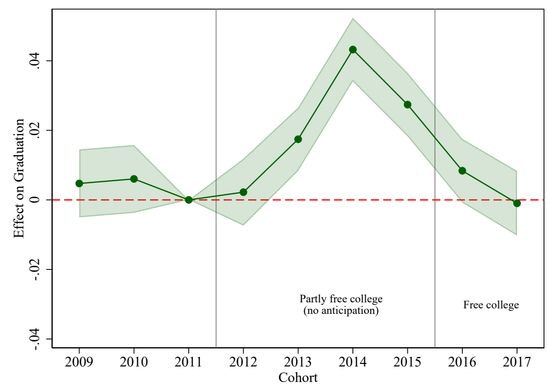
Figure 4 shows the results of estimating Equation 2. Panels a) to d) show the results for dropout, graduation, persistence, and graduation on time, respectively. We find that dropout decrease up to 2.5 p.p. for infra-marginal students, and the effect reverses for 2017 cohort. The composition of new entrants seem to dominate the positive effect from infra-marginal students. Graduation consequently also rises for students that receive the subsidy after their enrollment decision, while cohorts starting 2016 graduate at a simi-

[3] see Pal (2025) for a comprehensive analysis of this policy.

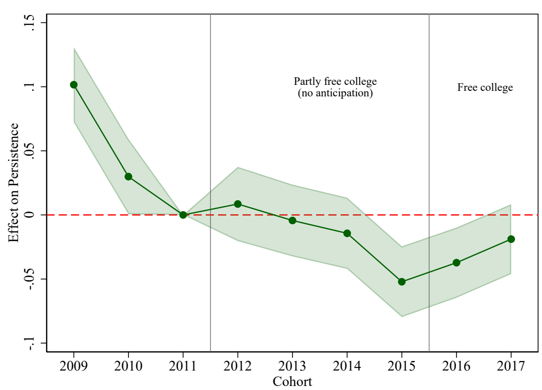
Figure 4: DID outcomes



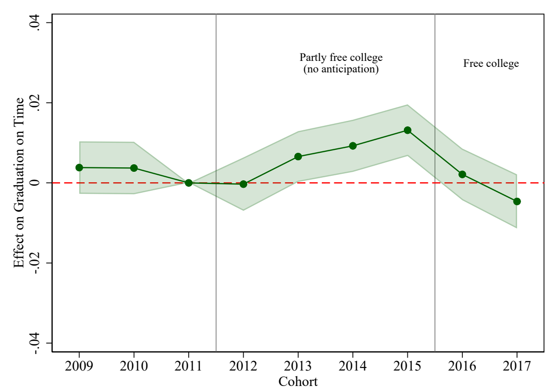
a) Dropout



b) Graduation



c) Persistence



d) Graduation on Time

Notes: These figures show the results of estimating Equation 1 on dropout, graduation, persistence, and graduation on time. Sample is enrolled students. Cohorts 2012 to 2015 are potentially eligible to TFHE starting 2016 but already made their initial enrollment decision. Cohort 2011 is the reference category.

lar rate as never-treated students. Persistence results are hard to interpret has the pre-treatment period exhibits a pronounced negative trend, which the effect of policy seems to moderate. Finally, as the subsidy gets larger (2015 cohorts), the proportion of students who graduates on time rises by 1.8 p.p., while there are null efects of TFHE starting 2016.

These results are defined as the overall effect of the policy on those who already enrolled. TFHE lowers the cost of staying enrolled, which tends to reduce dropout and increase graduation. It does so by lowering out-of-pocket and the removal of performance requirements to students who previously benefited from scholarships. Conversely, students with lower PSU scores that now enter higher education face more difficulties to progress and graduate. Quasi-experimental findings suggest that both forces are at play and cancel each other out. The total effect is unsignificantly different from zero, meaning the total number of graduates is larger (same graduation rate, but more students). To validate reduced-form results and quantify both effects, we next build a model of enrollment and major choice with endogenous effort.

3.4 Discussion and robustness

Our quasi-experimental analysis suggests that TFHE policy had a positive effect on enrollment on eligible students, especially for low-ability students that previously had limited access to public funding. We find no evidence of crowding-out of non-eligible institutions, suggesting that most of the increase in enrollment comes from marginal students. Eligible institutions were able to absorb the increase in enrollment, especially lower-quality institutions. Regarding educational outcomes, we find modest effects on dropout and graduation for infra-marginal students, while there are no significant effects for students that enrolled after the implementation of the policy, despite being of lower observed ability on average.

There are some limitations to our empirical strategy. We used eligibility to TFHE as a proxy for treatment, and compared eligible to non-eligible students as we do not have a continuous measure of income.

Crowd-out of average ability, high-income students If high-income students were to reduce their enrollment as a result of the policy, our estimates would be biased upwards. However, spillovers are unlikely as cut-offs scores were mostly unchanged, suggesting that the policy did not put additional stress on the capacity of institutions. As a robustness check, we estimate Equation 1 restricting the sample to income deciles 1, 2 and 9, 10, as crowd-out is more likely to occur around the eligibility threshold, and find similar results.

Increase in test-takers If the policy induced more low-income students to take the PSU test, our control group would be positively selected, biasing our results upwards. However, we find no evidence of an increase in test-takers as a result of the policy. This is because take-up rates were already high (around 85% of students in their last year of high school take the test). In fact, what we see is an increase in higher education applications from low income students.

Income manipulation Income is self-reported and verified by the goverment. Misreporting could lead to students that are actually ineligible being classified as eligible. If this is done on purpose, it could bias our estimates upwards.

4 Model

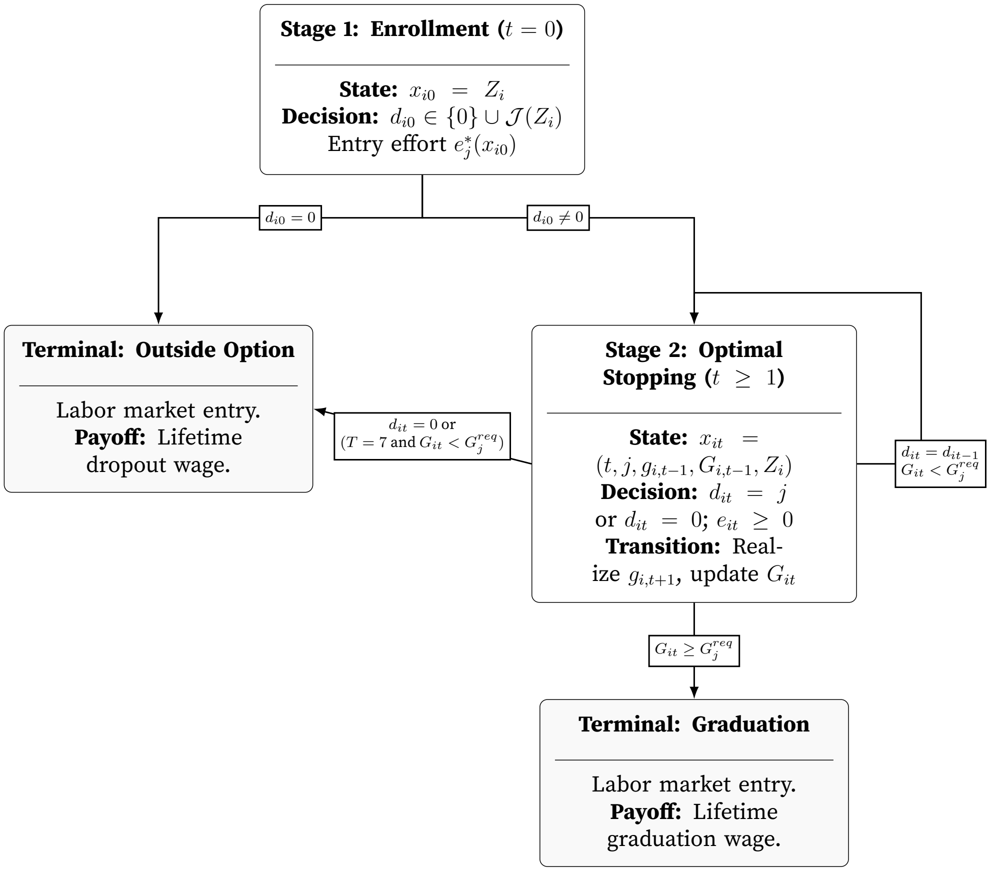
The decision process of an individual considering attending higher education can be depicted in two stages. In the first stage, she chooses whether to enroll, and if so, which field–institution combination to attend. To decide, she observes program characteristics and available funding instruments, and optimizes how much effort to exert in each program in her personalized choice set, determined by her score in the national exam (Fack et al., 2019). Effort choice will determine her distribution of performance outcomes (credits). Conditional on enrollment, performance today endogenously determines out-of-pocket fees for the subsequent year. The student graduates when she cumulates the required number of credits of the enrolled program.

Direct measures of effort are not observed. Instead, we attempt to uncover *effective* effort through realized outcomes, i.e. course credits completion (De Groote, 2025).

4.1 Timing and choices

Figure 5 describes the timing and choices of the model. A student is characterized by her type Z_i, defined by sex, ability (psu and gpa), and socioeconomic status. In stage 1, the student decides whether to enroll in higher education. She observes her type Z_i, which characterizes her personalized choice set $\mathcal{J}(Z_i)$ (Fack et al., 2019) and the funding instruments available. She decides between a field–institution combination $d_{i0} = j$, with $j \in \mathcal{J}(Z_i) = \{1, \ldots, J(Z_i)\}$ a program from her personalized choice set, and the outside option $j = 0$. The student solves for the optimal "entry effort" $e_j^*(x_{i0})$ for every potential program and chooses the one yielding the higher lifetime expected utility.

Figure 5: Structural model layout



Stage 2 is conditional on enrollment. It is an optimal stopping problem. The student decides in each subsequent period whether to continue in the program, $d_{it} = \{d_{i,t-1}, 0\}$, and how much effort $e_{it} \in [0, +\infty)$ to exert. By exerting effort, the student implicitly chooses the probabilities of performance outcomes $g_{i,t+1}$, which accumulates over time as $G_{it} := \sum_{s=1}^{t} g_{is}$.

The model is discrete in time, with $t \in \{0, 1, \ldots, T\}$ representing academic years. The nominal length of the program j is $\bar{t_j}$. The student can drop out at any time until the terminal period is $T = 7$ years, where she is forced to drop out if she did not yet graduate.

During periods before graduation ($t = 1, \ldots, \bar{t_j} - 1$), performance affects both the probability of dropout and, for scholarship holders, out-of-pocket fees $OP_j(x_{it})$: failing to meet the performance requirement ($g_{it} \geq 3$) implies paying the full tuition the following year. During the graduation period ($t = \bar{t_j}, \ldots, T$), the student can eventually graduate with some probability and obtain the continuation value associated with graduation, or alternatively postpone graduation if $t < T$, or drop out. At terminal period $T = 7$, any student who has not yet graduated is assumed to drop out. Hence, the state space is defined as

$x_{it} = (t, d_{i,t-1}, g_{i,t-1}, G_{i,t-1}, Z_i)$.

4.2 Utility and the dynamic program

Flow utility from being enrolled in program j during period t includes individual preferences for different programs, out-of-pocket fees sensitivity, and the psychological or physical "disutility" of academic work. We decompose it into a fixed cost (FC_j) that captures preferences for higher education, and a variable cost of effort:

$$u_j(x_{it}) + \varepsilon_{ijt} = -FC_j(x_{it}) - c_j(x_{it})e_{it} + \varepsilon_{ijt}, \tag{3}$$

with marginal cost $c_j(x_{it}) > 0$. Fixed costs $FC_j(x_{it})$ summarize all non-effort related preferences for programs including out-of-pocket fees $OP(Z_{i,}, g_{i,t-1}, P_{j,t})$, defined as the fraction of program fees not covered by the state for student i in program j. These depend on tuition $P_{j,t}$, personal characteristics Z_i and past performance $g_{i,t-1}$:

$$OP(Z_{i,}, g_{i,t-1}, P_{j,t}) = \begin{cases} (1 - \lambda(Z_i, g_{i,t-1}))P_{j,t} & \text{if the student holds a scholarship}, \\ (1 - \lambda(Z_i))P_{j,t} & \text{otherwise}. \end{cases}$$

where $\lambda(\cdot) \in [0, 1]$ is the subsidy rate, i.e. the fraction of program fees $P_{j,t}$ covered by the Government. For scholarship holders, $\lambda(\cdot)$ depends on past performance g_{it-1} as well as on individual characteristics Z_i, since failure to meet credit requirements results in the loss of the subsidy. For loans and TFHE, $\lambda(\cdot)$ depends only on observable characteristics and does not vary with performance. Finally, the distribution of the idiosyncratic shock $\varepsilon_{ijt} \sim$ EV1 is iid and common knowledge. It captures what the student learns at the beginning of the period [4].

The dynamics of the optimization problem are the following: On the one hand, by exerting more effort today, the student i) increases her probability of graduating in subsequent periods and thereby raises her expected future wage, and ii) increases her probability to maintain public funding (passing the performance threshold) if eligible. On the other hand, exerting effort is costly and reduces the flow utility of attending higher education. The conditional value function $v_j(x_{it}, e_{it})$ can be written as

$$v_j(x_{it}, e_{it}) + \varepsilon_{ijt} = u_j(x_{it}, e_{it}) + \beta \sum_{\bar{g} \in \mathcal{G}} \phi^{\bar{g}}(e_{it}) \bar{V}(x_{i,t+1}(\bar{g})) + \varepsilon_{ijt}, \tag{4}$$

[4]

where $\bar{V}(x_{i,t+1}(\bar{g}))$ denotes the ex-ante value function conditional on choosing program j. The first term corresponds to the current flow utility from enrolling in field–institution j. The second term captures the expected continuation value, discounted by β. Uncertainty in $x_{i,t+1}(\bar{g})$ arises solely from the realization of the performance outcome $\bar{g}$.

Applying the log-sum expression to Equation 4 yields

$$v_j(x_{it}, e_{it}) + \varepsilon_{ijt} = u_j(x_{it}, e_{it}) + \beta \sum_{\bar{g} \in \mathcal{G}} \phi^{\bar{g}}(e_{it}) \ln\left(\sum_{j \in \mathcal{J}} \exp\big(v_j(x_{i,t+1}, e_{i,t+1})\big) \right) + \beta\gamma + \varepsilon_{ijt}.$$

4.3 *Effective* effort and performance

Effective effort e_{it} is unobserved, while performance outcomes g_{it} are directly recovered from students' records. We model performance outcome $g_{i,t+1}$ as the result of effort e_{it} and a logistically distributed shock η_{it}:

$$g_{i,t+1} = \kappa \quad \text{if } \bar{g}^{\kappa}_{jt} < \ln(e_{it}) + \eta_{i,t+1} \leq \bar{g}^{\kappa+1}_{jt}, \tag{5}$$

where the cutoff values $\bar{g}^{\kappa}_{jt}$ and $\bar{g}^{\kappa+1}_{jt}$ are field–institution–year specific. In practice, performance is discretized into five categories, $\kappa \in \{0, 1, 2, 3, 4\}$. Students observe the realization of the shock $\eta_{i,t+1}$ only after choosing their effort e_{it}, but since they know the distribution of $\eta_{i,t+1}$ (standard logistic), they can compute the probability of each outcome conditional on effort:

$$\Pr\left(g_{i,t+1} = \kappa \mid e_{it}, x_{it}\right) = F\big(\ln(e_{it}) - \bar{g}^{\kappa}_{jt}\big) - F\big(\ln(e_{it}) - \bar{g}^{\kappa+1}_{jt}\big), \tag{6}$$

where $F(\cdot)$ denotes the logistic CDF. Making use of the logistic assumption of the performance shock $\eta_{i,t+1}$, for $\kappa = 0$ we can rewrite Equation 6 as:

$$e_{it} = \frac{1 - \Pr(g_{i,t+1} = 0 \mid e_{it}, x_{it})}{\Pr(g_{i,t+1} = 0 \mid e_{it}, x_{it})}. \tag{7}$$

The higher the effort, the higher the expected performance. By choosing effort, students choose the distribution of performance outcomes. Effective effort can thus be interpreted as the odds of avoiding the lowest performance outcome (failing all registered credits).

4.4 Solution of the model

The model is solved by backward induction. Higher education is no longer feasible after seven years, or once the student completes all program course credits. Formally,

$$\bar{V}(x_{i,t+1}(\bar{g})) = \alpha_w \, wage_j(x_{i,t+1}) \qquad \text{if } t+1 = T = 7 \text{ or } G_{i,t+1} \geq G_j^{\text{req}}. \tag{8}$$

and is used as an input in earlier periods. Here, $wage_j(x_{i,t+1})$ denotes lifetime expected wage for program j or dropout wage if $d_{i,t+1} = 0$.

At each period t, the student first chooses an effort level e_{it} in her program j. The decision trades off a loss in flow utility with higher future expected gains. Assuming she is rational and behaving optimally, she is solving the following first-order condition for each program [5]:

$$\frac{\partial v_j(x_{it}, e_{it})}{\partial e_{it}} = \frac{\partial u_j(x_{it}, e_{it})}{\partial e_{it}} + \beta \sum_{g \in \mathcal{G}} \frac{\partial \phi^{\bar{g}}(e_{it})}{\partial e_{it}} \bar{V}(x_{it+1}(\bar{g})) = 0 \quad \text{if } e_{it} = e_j^*(x_{it}). \tag{9}$$

Given the set of optimal effort levels $\{e_j^*(x_{it})\}_{j \in \mathcal{J}(x_{it})}$, the student chooses the program with the highest value $v_j(x_{it}, e_j^*(x_{it}))$. The resulting choice probabilities take the familiar logit form:

$$p_{jt} = \frac{\exp\big(v_j(x_{it}, e_j^*(x_{it}))\big)}{\sum_{j' \in \mathcal{J}(x_{it})} \exp\big(v_{j'}(x_{it}, e_{j'}^*(x_{it}))\big)}.$$

The ex-ante value function in period t can again be expressed, using the logsum formula, as

$$\bar{V}(x_{it}) = \gamma + \ln\left(\sum_{j \in \mathcal{J}(x_{it})} \exp\big(v_j(x_{it}, e_j^*(x_{it}))\big) \right), \tag{10}$$

and the model is solved recursively by iterating backward to $t = 1$.

[5] a sufficient condition for an interior solution is that students believe there is a non-zero probability to avoid the lowest performance outcome. Positive marginal costs ensure an infinite amount of effort is never exerted.

4.5 Identification

We set the discount factor β to 0.95 and normalize the utility of the outside option to the dropout wage (Magnac & Thesmar, 2002).

$$\begin{cases} u_j(x_{it}, e_{it}) = -FC_j(x_{it}) - c_j(x_{it})e_{it}, & \text{if } j \neq 0, \\ u_0(x_{it}) = \alpha_w wage_0(x_{it}), & \text{if } j = 0. \end{cases} \tag{11}$$

We assume that not enrolling in higher education (and dropping out) is a terminal action. This means we do not allow for individuals to resit the exam one year after or re-enter higher education after dropping out.

4.5.1 Fixed and Marginal costs

Marginal costs c_j are identified from the FOC (De Groote, 2025). The model attributes higher performance levels, conditional on the same continuation value, to lower marginal costs of effort. Fixed costs then rationalize the observed enrollment shares across programs. The sensitivity to out-of-pocket fees is pinned down by the variation in public funding eligibility, after controlling for observables, which serves as a price shifter.

Wage elasticity α_w is identified through an exclusion restriction. We use local labor market field-region variation. The underlying assumption is that conditional on region and field fixed effects, the interaction provides idiosyncratic local demand shocks that shift utility only through the wage.

4.5.2 Effective effort

We assume students are rational and adjust effective effort so as to maxmize expected lifetime utility. As discussed in Section 4.3, it is a tool to recover the distribution of performance outcomes conditional on state variables. We specify it as the index of a flexible ordered logit model of credit performance, where credits are discretized into five categories (0-4). The exact specification can be found in Appendix C. Performance thresholds are recovered from this estimation, and are allowed to vary by field, institution and year.

4.5.3 Decomposing adverse selection and moral hazard

We would like to decompose the effects of the policy into two components: adverse selection and moral hazard. First, by lowering out-of-pocket costs, it predicts new students to enroll. Second, by altering the consequences of academic performance while enrolled, it may change effort provision and therefore performance.

Let $r \in \{1, ..., R\}$ denote the policy regime. Define $D_i^r \in \{0, 1\}$ as the enrollment decision of student i under regime r. We call

$$\mathcal{M} = \{i : D_i^r \neq D_i^{r'}\}$$

the set of marginal students, i.e. those whose enrollment decision changes because of the policy. This group captures the extensive-margin composition effect.

Define instead

$$\mathcal{IM} = \{i : D_i^r = D_i^{r'} = 1\}$$

as the set of infra-marginal students, who enroll regardless of the policy regime. For these students, changes in outcomes reflect only the change in post-enrollment incentives. Let Y_{it}^r denote the potential performance outcome of student i in period t under regime r. Then the moral-hazard effect is identified from

$$\mathbb{E}\left[Y_{it}^r - Y_{it}^{r'} \mid i \in \mathcal{IM}\right].$$

In the structural model, these outcomes can be written as

$$Y_{it}^r = Y_{it}\big(FC_j^r(x_{it}), c_j^r(x_{it}), e_j^r(x_{it})\big),$$

so that moral hazard is the change in performance among students whose enrollment decision is unchanged, while adverse selection is captured by the characteristics and outcomes of the marginal entrants $\mathcal{M}$.

4.6 Estimation

We avoid full solution methods by representing the future value term as the conditional value function and a non-negative term that depends on the empirical choice probabilities (Hotz & Miller, 1993). Given the Type-1 extreme value assumption on utility shocks, CCPs are of logit type. We specify flexible parametric functional forms for CCPs, effective effort, fixed costs and wage regression. Details on the estimation of these components are provided in Appendix C.

Exiting higher education is a terminal action in the model, so its conditional value function of dropping out equals flow utility, i.e. dropout wage. It is therefore convenient to use $j' = 0$ as an arbitrary choice and write

$$\begin{aligned}
v_j(x_{it}, e_{it}) + \varepsilon_{ijt} &= u_j(x_{it}, e_{it}) \\
&\quad + \beta \sum_{\bar{g} \in \mathcal{G}} \phi^{\bar{g}}(e_{it}) \ln \left(\sum_{j' \in \mathcal{J}} \exp\big(v_{j'}(x_{i,t+1}(\bar{g}))\big) \right) + \beta\gamma + \varepsilon_{ijt} \\
&= u_j(x_{it}, e_{it}) \\
&\quad + \beta \sum_{\bar{g} \in \mathcal{G}} \phi^{\bar{g}}(e_{it}) \left(\alpha_w wage_0(x_{i,t+1}) - \ln p_{0,t+1}(x_{i,t+1}(\bar{g})) \right) + \beta\gamma + \varepsilon_{ijt},
\end{aligned}$$

by substituting $v_0(x_{i,t+1}(\bar{g})) = \alpha_w wage_0(x_{i,t+1})$ and exploiting dropping out as a terminal action. $p_{0,t+1}$ is the step-ahead dropout probability and γ the Euler-Mascheroni constant. Note that the ex-ante value function is now a function of performance probabilities, the dropout wage and the step-ahead dropout probability, which can all be estimated from the data. From the first-order condition in Equation 9, we can directly recover the expression for marginal costs.

$$c_j(x_{it}) = \beta \sum_{\bar{g} \in \mathcal{G}} \frac{\partial \phi^{\bar{g}}(e_{it})}{\partial e_{it}} \bar{V}(x_{it+1}(\bar{g})) \quad \text{if } e_{it} = e_j^*(x_{it}), \tag{12}$$

where $\bar{V}(x_{it+1}(\bar{g}))$ is computed using the estimated CCPs and dropout wage, and $\frac{\partial \phi^{\bar{g}}(e_{it})}{\partial e_{it}}$ can be directly computed given the logistic assumption about η_{t+1} in Equation 5.

Estimation can be performed following three sequential steps.

1. We flexibly estimate $p_{0,t+1}$ from a logistic regression and $\ln e_j^*(x_{it})$ the underlying index of an ordered logit model of credit completion (0-4).

2. The FOC equates marginal cost to marginal benefits, with an expression that can be directly estimated from the data. We can substitute the expression in $v_j(x_{it})$ and estimate $FC_j(x_{it})$ by maximum likelihood.

3. Finally, we use the estimates and the FOC to compute $c_j(x_{it})$.

5 Results

This Section first presents reduced-form results from dropout probabilities and performance regressions. It then discusses the structural parameters, which include students' fixed and marginal costs of effort. Taken together, these estimates pin down how students respond to changes in out-of-pocket tuition and to the presence or absence of performance requirements, and underlie the decomposition of the policy's aggregate effects into a selection and a behavioral component.

5.1 Dropout probabilities

Table 1: Dropout probabilities

	dropout	
OP fees	0.044***	(0.003)
Rel. Ability	0.554***	(0.014)
Distance	0.041***	(0.015)
Cumulated credits (t-1)	-0.217***	(0.003)
Credits (t-1)	-0.561***	(0.006)
Delay (t-1)	0.482***	(0.019)
female	-0.487***	(0.040)
Middle SES	0.628***	(0.048)
High SES	-1.367***	(0.052)
PSU	-1.021***	(0.038)
GPA	-0.120***	(0.024)
OP $\times$ female	0.002	(0.003)
OP $\times$ middle SES	-0.085***	(0.004)
OP $\times$ high SES	0.105***	(0.005)
OP $\times$ psu	-0.024***	(0.002)
OP $\times$ gpa	0.001	(0.002)
Delay (t-1) $\times$ female	0.560***	(0.020)
Delay (t-1) $\times$ middle SES	-0.184***	(0.024)
Delay (t-1) $\times$ high SES	0.596***	(0.027)
Delay (t-1) $\times$ psu	-0.239***	(0.013)
Delay (t-1) $\times$ gpa	0.130***	(0.013)
Cumul (t-1) $\times$ female	-0.019***	(0.002)
Cumul (t-1) $\times$ middle SES	-0.017***	(0.002)
Cumul (t-1) $\times$ high SES	0.038***	(0.002)
Cumul (t-1) $\times$ psu	0.017***	(0.001)
Cumul (t-1) $\times$ gpa	0.004***	(0.001)
Credits (t-1) $\times$ female	0.070***	(0.006)
Credits (t-1) $\times$ middle SES	0.037***	(0.008)
Credits (t-1) $\times$ high SES	-0.023***	(0.008)
Credits (t-1) $\times$ psu	-0.002	(0.004)
Credits (t-1) $\times$ gpa	0.038***	(0.004)
t= 2	-0.481***	(0.013)
t= 3	-0.419***	(0.019)
t= 4	-0.321***	(0.026)
t= 5	0.219***	(0.032)
t= 6	1.881***	(0.037)
t= 7	4.549***	(0.050)
Observations	1,098,085	
Mean of Dep. Variable	0.218	

Standard errors in parentheses

Logistic regression of choosing the outside option (dropping out) in period t. Controls include vector of characteristics, Institution and major FE, year enrolled FE and cumulated performance. PSU and GPU are standardised. Base category is low SES and t=1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 1 reports the results of a logistic regression model of dropout. Overall, dropout probabilities depend strongly on the enrollment length. Progressing through the program reduces the likelihood of dropping out (relative to period 1), although the probability rises again when students reach the nominal length of the degree. Being female reduces the probability of dropping out, with the effect becoming stronger as students progress in their studies.

Ability is strongly correlated with persistence in higher education, but conditional on the student's progress (accumulated credits), its estimated coefficient diminishes. Both cumulative credits and the number of credits completed in the previous year are strong determinants of student progression. Being delayed in the program increases the probability of dropping out, with a stronger magnitude for females.

5.2 Performance

Table B3 presents the results of the ordered logit model of performance with five categories (0–4). Coefficients follow a similar pattern as in the logistic dropout model. Ability is a strong predictor of performance, as is income. Out-of-pocket fees have a negative effect on performance, which is stronger for low-income and low-ability students.

Being delayed in the program reduces the probability of achieving the maximum number of credits, although the effect is milder for females. Finally, past progression, measured through cumulative and previous-year credits, is a strong predictor of current performance, particularly for low-income students.

5.3 Fixed Costs

Estimates of the conditional value function are presented in Table 2. Students' fixed costs increase with out-of-pocket fees, particularly at the program choice stage. Once enrolled, this effect is substantially attenuated in the continuation decision. Females are more sensitive to price than males. Students also face higher fixed costs when the program is located in a different region.

Past progression significantly affects fixed costs, as reflected in the coefficient of the delay variable. Again, females face higher fixed costs than males when delayed, and students who performed well on the national exam appear to experience smaller penalties from lagging behind. Expected wage has a positive effect on the conditional value function, indicating that students value their future expected earnings.

Table 2: Fixed costs

	cvf	
OP fees	-0.588***	(0.037)
Distance	-1.397***	(0.031)
Delay (t-1)	-2.744***	(0.127)
OP x female	-0.106***	(0.034)
OP x middle SES	0.026	(0.040)
OP x high SES	-0.056	(0.045)
OP x psu	0.014	(0.022)
OP x gpa	0.015	(0.020)
OP (during HE)	0.440***	(0.042)
Delay (t-1) x female	-0.280**	(0.136)
Delay (t-1) x middle SES	-0.150	(0.160)
Delay (t-1) x high SES	-0.291	(0.181)
Delay (t-1) x psu	0.377***	(0.075)
Delay (t-1) x gpa	-0.131	(0.086)
EMAX	0.950	(.)
Expected wage	0.001***	(0.000)
Field FE	Yes	
Institution FE	Yes	

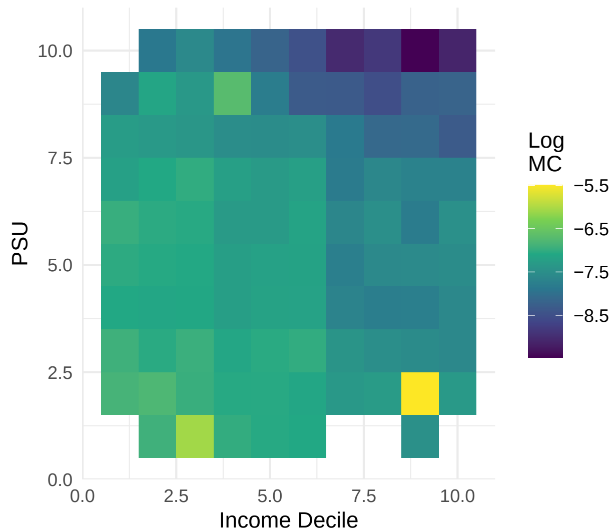
Standard errors in parentheses

Conditional Value function Estimation

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5.4 Marginal Costs

Figure 6: Distribution of marginal costs, by income decile and PSU score



Notes: Heatmap of log marginal costs by PSU (y-axis) and income (x-axis) decile.

Figure 6 shows the distribution of log marginal costs by income and PSU score. The heatmap exhibits a clear gradient in marginal costs along both ability and socioeconomic status dimensions, with low-PSU and low-income students facing systematically higher marginal costs of producing a unit of academic progress.

Table 3 presents, for interpretational purposes, an OLS regression of log marginal costs on state variables. Marginal costs decrease with out-of-pocket fees, in particular for low-income students. Females tend to have higher marginal costs compared to males, especially at lower income levels.

5.5 Selection versus behavioral responses

Taken together, the structural estimates speak directly to the two forces at work in the post-reform period. Marginal entrants induced by *Gratuidad*, predominantly lower-ability and lower-income students, exhibit both higher fixed costs of enrollment and higher marginal costs of effort. They are therefore predicted to progress more slowly and to drop out more often, consistent with the leftward shift in the ability distribution of post-2016 cohorts.

Among infra-marginal students, by contrast, the reduction in out-of-pocket fees lowers fixed costs and, through the removal of the threat of losing the scholarship, raises the conditional value of continuation. The estimated marginal costs of effort do not increase

Table 3: Marginal costs

	log(mc)	
female	-0.226**	(0.105)
Middle SES	-0.005	(0.108)
High SES	0.242**	(0.115)
Standardized values of psu_avg	-0.322***	(0.120)
Standardized values of gpa_hs	-0.130**	(0.058)
OP fees	-0.038**	(0.019)
Distance	0.064***	(0.025)
Delay (t-1)	-0.801***	(0.170)
t= 2	-0.221***	(0.075)
t= 3	-0.107	(0.079)
t= 4	-0.039	(0.097)
t= 5	2.393***	(0.089)
t= 6	2.932***	(0.180)
t= 7	4.966***	(0.177)
OP x female	0.032	(0.020)
OP x middle SES	-0.017	(0.018)
OP x high SES	-0.018	(0.026)
OP x psu	-0.009	(0.010)
OP x gpa	0.007	(0.012)
Delay (t-1) x female	0.158	(0.127)
Delay (t-1) x middle SES	0.421***	(0.151)
Delay (t-1) x high SES	0.307*	(0.166)
Delay (t-1) x psu	0.034	(0.074)
Delay (t-1) x gpa	-0.011	(0.070)
Observations	10,811	
ymean		

Standard errors in parentheses

MC heterogeneity

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

in response to the elimination of performance requirements, so there is no detectable moral-hazard response to the removal of the conditionality. The model therefore rationalizes the reduced-form evidence reported in Section 3. The two forces, namely adverse selection on costs among new entrants and modest improvements in incentives among incumbents, approximately offset in the aggregate, leaving completion rates roughly unchanged while increasing the absolute number of graduates.

6 Conclusion

This paper investigated the impacts of Tuition-Free Higher Education on student enrollment, persistence, and academic performance, using Chile's 2016 *Gratuidad* reform as a natural experiment. Our difference-in-differences analysis reveals that the policy boosted enrollment among eligible students by around 10 p.p., with the response concentrated among lower-ability students who, prior to the reform, did not qualify for either subsidized loans or merit scholarships. We find no evidence of crowding-out at non-eligible institutions, suggesting that the bulk of the response comes from genuinely marginal students. Holding composition fixed using cohorts that had already enrolled before 2016, dropout fell by up to 2.5 p.p. and on-time graduation rose by 1.8 p.p. among infra-marginal students. For post-reform cohorts, aggregate completion rates remained essentially unchanged despite the negative composition shift.

To unpack these aggregate effects, we developed and estimated a dynamic discrete-choice model of enrollment, program choice, and academic effort in which students respond to changes in out-of-pocket tuition and to the presence or absence of performance requirements. The estimates reveal substantial heterogeneity in fixed and marginal costs across the student distribution, with low-ability and low-income students facing systematically higher marginal costs of producing a unit of academic progress. Critically, we find no evidence that the removal of performance requirements weakened effort among infra-marginal students. Instead, the composition of new entrants induced by *Gratuidad* accounts for most of the post-reform progression patterns. The model therefore rationalizes the reduced-form evidence: aggregate completion is held in check by composition rather than by moral hazard.

These findings have implications for higher education financing more broadly. They suggest that unconditional aid can promote access without compromising aggregate educational quality, especially in contexts with high tuition costs and imperfect income verification. At the same time, the size of the composition effect indicates that equity gains come together with a larger share of students who struggle to progress, with corresponding implications for the public budget and for the average return per dollar spent.

In ongoing work, we use the estimated model to simulate alternative funding designs that re-introduce ability or performance-based conditions on top of the tuition-free benefit, and assess their implications for fiscal cost and educational attainment. These counterfactual exercises will allow us to evaluate budget-improving alternatives that preserve the equity gains of *Gratuidad*, and to speak to a wider design question facing governments that have already invested in tuition-free systems.

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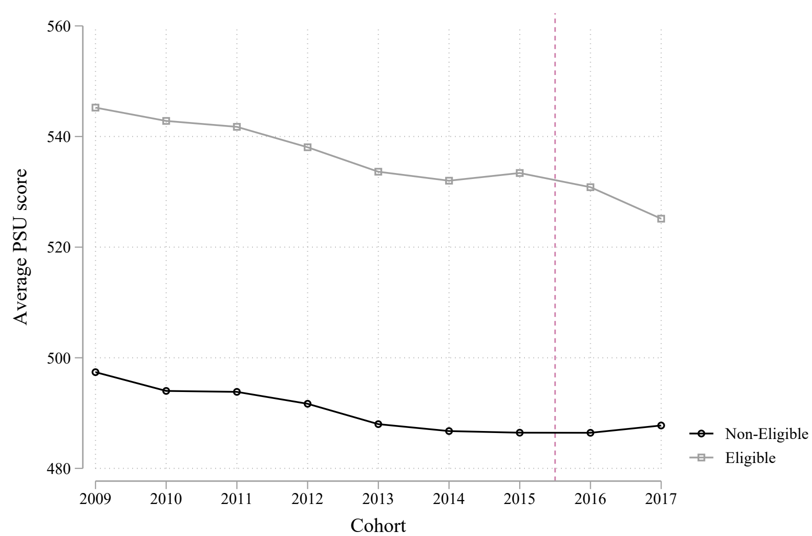
Appendices

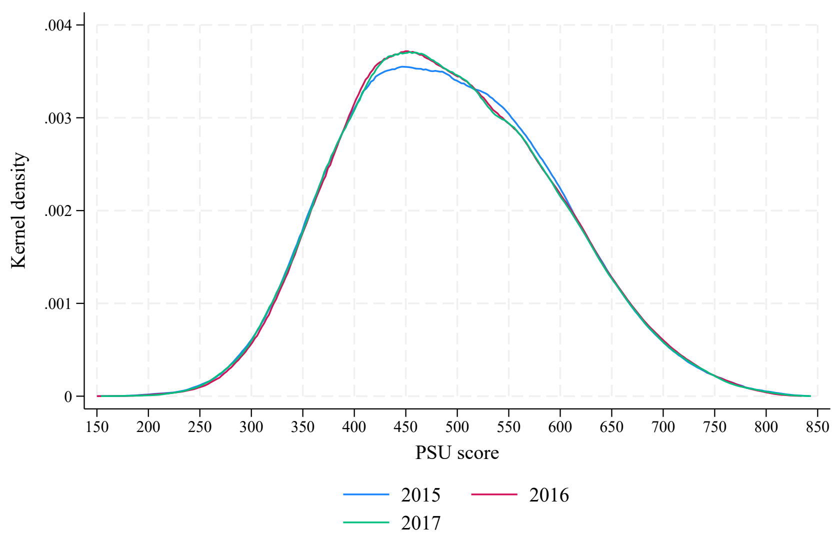
A Figures

Figure A1: Average PSU score over time for participating and non-participating institutions



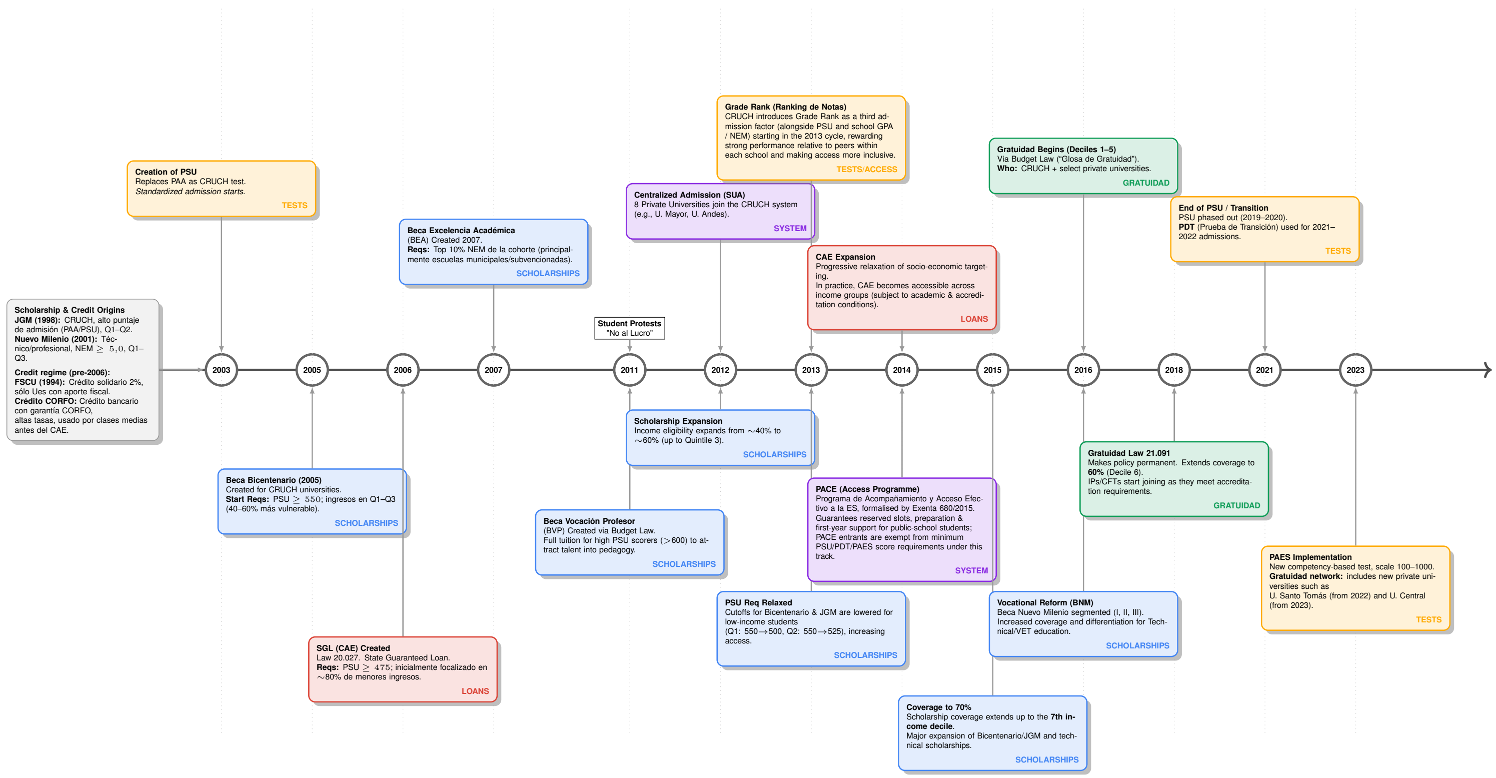
Notes: Annual average PSU score by groups of eligible and non-eligible instutions to free higher education.

Figure A2: Average PSU score over time for participating and non-participating institutions



Notes: Kernel density of PSU average scores enrolled in higher education for years 2015, 2016 and 2017.

Figure A3: Education reforms 2003-2024



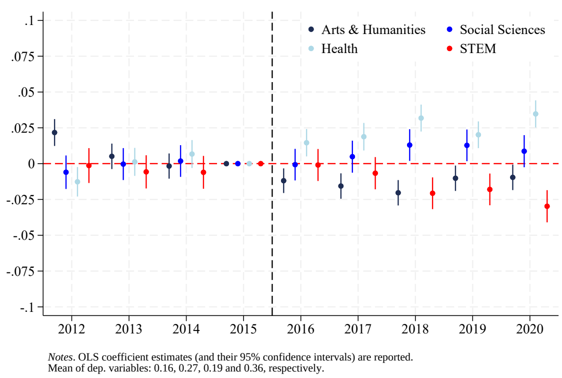
	PSU ≥ 510	$475 \leq$ **PSU** < 510	**PSU** < 475
Deciles 1-5	**Scholarship** **TFHE (univ)** **TFHE (univ+SCP)**	**SGL loan** **TFHE (univ)** **TFHE (univ+SCP)**	**No public aid** **TFHE (univ)** **TFHE (univ+SCP)**
Deciles 6-7	**Scholarship**	**SGL loan**	**No public aid**
Deciles 8-10	**SGL loan**	**SGL loan**	**No public aid**

PSU = admissions test; SGL = State-Guaranteed Loan; SCP = short-cycle programs; TFHE = Tuition-Free Higher Education

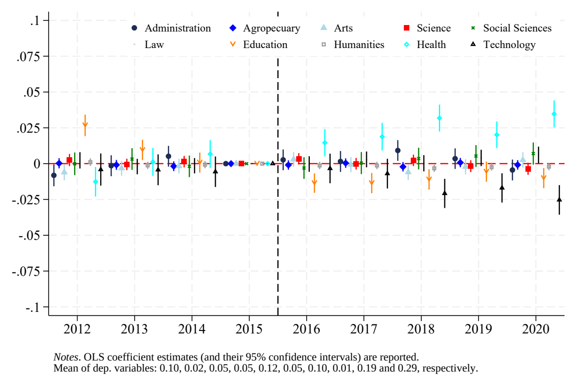
Years: **2015** **2016** **2017**

Figure A4: Changes in funding instruments after TFHE implementation

Figure A5: DID: Preferences



a) 4 fields



b) 10 fields

Notes: These figures show the results of estimating Equation 1. Panel (a) groups the results in four fields, while panel (b) uses the original ten-field classification. STEM includes Agropecuary, Science and Technology. Social Sciences includes Administration, Social Sciences nad Law. Arts and Humanities includes Education, Arts and Humanities. Health includes Health Sciences.

B Tables

Table B1: Descriptive statistics, students

	2012	2013	2014	2015	2016	2017
Enrollment						
N Students	165762	169257	169415	176027	180128	180302
Enrolled in platform	.260	.273	.280	.275	.280	.276
Enrolled out of platform	.338	.351	.346	.341	.330	.329
Not enrolled	.400	.374	.372	.382	.388	.394
Demograhics						
Family Income	3.5	3.7	3.9	4	4.1	4.5
Private School	.111	.111	.111	.109	.106	.105
Private Health	.268	.267	.268	.263	.262	.235
Father With College	.168	.169	.170	.171	.169	.184
Mother Employed	.414	.436	.460	.460	.461	.461
Test Score	490	491	491	492	492	491
Free College	0	0	0	0	.145	.243
Subsidized Loan	.222	.213	.215	.187	.145	.105
Merit-based Scholarship	.136	.191	.200	.249	.158	.101
Field						
Business	.136	.136	.139	.148	.155	.160
Farming	.021	.022	.021	.023	.024	.025
Art and Architecture	.051	.050	.051	.052	.053	.055
Basic Sciences	.032	.033	.035	.033	.034	.031
Social Sciences	.081	.077	.076	.078	.083	.083
Law	.038	.035	.037	.037	.040	.040
Education	.110	.096	.091	.091	.095	.095
Humanities	.010	.010	.010	.010	.010	.009
Health	.214	.201	.200	.196	.196	.196
Technology	.284	.318	.318	.312	.290	.289

Notes: This table shows descriptive statistics on every student who enrolled and took the college entrance exam. Family income is categorized in 1-10 brackets, and field clasification is performed following the ISCED-UNESCO guidelines.

Table B2: Dropout Probabilities

	dropout	
OP fees	0.044***	(0.003)
Rel. Ability	0.554***	(0.014)
Distance	0.041***	(0.015)
Cumulated credits (t-1)	-0.217***	(0.003)
Credits (t-1)	-0.561***	(0.006)
1 left (t-1)	3.659***	(0.032)
2 left (t-1)	2.574***	(0.042)
3 left (t-1)	1.959***	(0.049)
Delay (t-1)	0.482***	(0.019)
female	-0.487***	(0.040)
Middle SES	0.628***	(0.048)
High SES	-1.367***	(0.052)
PSU	-1.021***	(0.038)
GPA	-0.120***	(0.024)
OP × female	0.002	(0.003)
OP × middle SES	-0.085***	(0.004)
OP × high SES	0.105***	(0.005)
OP × psu	-0.024***	(0.002)
OP × gpa	0.001	(0.002)
1 left (t-1) × female	0.644***	(0.035)
1 left (t-1) × middle SES	-0.157***	(0.047)
1 left (t-1) × high SES	0.420***	(0.043)
1 left (t-1) × psu	-0.218***	(0.023)
1 left (t-1) × gpa	-0.014	(0.022)
2 left (t-1) × female	0.591***	(0.048)
2 left (t-1) × middle SES	-0.206***	(0.064)
2 left (t-1) × high SES	0.388***	(0.058)
2 left (t-1) × psu	0.022	(0.031)
2 left (t-1) × gpa	0.125***	(0.030)
3 left (t-1) × female	0.693***	(0.055)
3 left (t-1) × middle SES	-0.098	(0.073)
3 left (t-1) × high SES	0.421***	(0.066)
3 left (t-1) × psu	-0.107***	(0.037)
3 left (t-1) × gpa	0.210***	(0.034)
Delay (t-1) × female	0.560***	(0.020)
Delay (t-1) × middle SES	-0.184***	(0.024)
Delay (t-1) × high SES	0.596***	(0.027)
Delay (t-1) × psu	-0.239***	(0.013)
Delay (t-1) × gpa	0.130***	(0.013)
t=2 × SCP	0.577***	(0.018)
t=3 × SCP	-0.105***	(0.024)
t=4 × SCP	0.362***	(0.033)
t=5 × SCP	-0.325***	(0.046)
t=6 × SCP	-1.002***	(0.065)
t=7 × SCP	-1.207***	(0.177)

Cumul (t-1) × female	-0.019***	(0.002)
Cumul (t-1) × middle SES	-0.017***	(0.002)
Cumul (t-1) × high SES	0.038***	(0.002)
Cumul (t-1) × psu	0.017***	(0.001)
Cumul (t-1) × gpa	0.004***	(0.001)
Credits (t-1) × female	0.070***	(0.006)
Credits (t-1) × middle SES	0.037***	(0.008)
Credits (t-1) × high SES	-0.023***	(0.008)
Credits (t-1) × psu	-0.002	(0.004)
Credits (t-1) × gpa	0.038***	(0.004)
Rel. Abil. x female	0.044***	(0.007)
Rel. Abil. x middle SES	-0.060***	(0.008)
Rel. Abil. x high SES	0.031***	(0.009)
Rel. Abil. x psu	-0.014***	(0.004)
Rel. Abil. x gpa	0.010***	(0.004)
Dist. x female	0.031*	(0.017)
Dist. x middle SES	0.048**	(0.021)
Dist. x high SES	0.005	(0.021)
Dist. x psu	0.026**	(0.011)
Dist. x gpa	-0.036***	(0.010)
Agricultural × fem	-0.100*	(0.055)
Agricultural × middle SES	0.092	(0.067)
Agricultural × high SES	-0.401***	(0.066)
Agricultural × psu	-0.066	(0.044)
Agricultural × gpa	0.081**	(0.034)
Arts and Architecture × fem	0.099**	(0.044)
Arts and Architecture × middle SES	0.028	(0.056)
Arts and Architecture × high SES	-0.227***	(0.053)
Arts and Architecture × psu	0.030	(0.036)
Arts and Architecture × gpa	-0.005	(0.028)
Science × fem	0.014	(0.046)
Science × middle SES	0.132**	(0.058)
Science × high SES	-0.017	(0.054)
Science × psu	0.302***	(0.037)
Science × gpa	0.222***	(0.029)
Social Sciences × fem	-0.030	(0.037)
Social Sciences × middle SES	-0.097**	(0.045)
Social Sciences × high SES	0.084*	(0.045)
Social Sciences × psu	0.171***	(0.029)
Social Sciences × gpa	0.016	(0.023)
Law × fem	-0.031	(0.041)
Law × middle SES	0.205***	(0.051)
Law × high SES	-0.297***	(0.048)
Law × psu	0.117***	(0.032)
Law × gpa	0.072***	(0.025)
Education × fem	-0.093**	(0.039)
Education × middle SES	-0.143***	(0.045)

Education × high SES	0.186***	(0.049)
Education × psu	0.108***	(0.030)
Education × gpa	-0.040*	(0.023)
Humanities × fem	-0.007	(0.098)
Humanities × middle SES	-0.085	(0.109)
Humanities × high SES	0.126	(0.115)
Humanities × psu	-0.149*	(0.079)
Humanities × gpa	0.102*	(0.060)
SCP × fem	-0.160***	(0.057)
SCP × middle SES	-0.117*	(0.065)
SCP × high SES	0.194**	(0.082)
SCP × psu	0.328***	(0.041)
SCP × gpa	-0.161***	(0.034)
SCP-Gratuidad × fem	-0.129***	(0.030)
SCP-Gratuidad × middle SES	-0.097***	(0.036)
SCP-Gratuidad × high SES	0.262***	(0.040)
SCP-Gratuidad × psu	0.290***	(0.025)
SCP-Gratuidad × gpa	-0.092***	(0.019)
Health × fem	-0.030	(0.034)
Health × middle SES	0.068*	(0.040)
Health × high SES	-0.144***	(0.040)
Health × psu	0.432***	(0.026)
Health × gpa	0.031	(0.020)
Technology × fem	-0.053	(0.033)
Technology × middle SES	0.060	(0.039)
Technology × high SES	-0.152***	(0.038)
Technology × psu	0.356***	(0.025)
Technology × gpa	0.166***	(0.019)
t= 2	-0.481***	(0.013)
t= 3	-0.419***	(0.019)
t= 4	-0.321***	(0.026)
t= 5	0.219***	(0.032)
t= 6	1.881***	(0.037)
t= 7	4.549***	(0.050)
Agricultural	-0.130**	(0.053)
Arts and Architecture	0.344***	(0.043)
Science	0.250***	(0.045)
Social Sciences	0.137***	(0.035)
Law	0.307***	(0.039)
Education	0.136***	(0.035)
Humanities	0.641***	(0.096)
SCP	0.545***	(0.077)
SCP-Gratuidad	0.085*	(0.043)
Health	0.113***	(0.033)
Technology	0.314***	(0.029)
mineduc=2	1.151***	(0.051)
mineduc=3	1.233***	(0.054)

mineduc=4	0.485***	(0.044)
mineduc=9	0.773***	(0.056)
mineduc=10	0.835***	(0.045)
mineduc=11	1.284***	(0.084)
mineduc=13	1.508***	(0.049)
mineduc=20	1.560***	(0.044)
mineduc=22	-0.249***	(0.045)
mineduc=23	1.471***	(0.066)
mineduc=31	0.731***	(0.039)
mineduc=34	1.985***	(0.067)
mineduc=39	0.832***	(0.039)
mineduc=42	0.732***	(0.051)
mineduc=45	0.806***	(0.050)
mineduc=50	0.204***	(0.051)
mineduc=54	0.487***	(0.041)
mineduc=69	1.329***	(0.055)
mineduc=70	1.948***	(0.059)
mineduc=71	1.100***	(0.050)
mineduc=72	1.059***	(0.047)
mineduc=73	0.590***	(0.048)
mineduc=74	0.765***	(0.052)
mineduc=75	0.541***	(0.050)
mineduc=76	0.819***	(0.050)
mineduc=77	0.697***	(0.059)
mineduc=78	0.964***	(0.051)
mineduc=79	0.462***	(0.051)
mineduc=80	0.476***	(0.046)
mineduc=81	0.682***	(0.053)
mineduc=82	1.326***	(0.063)
mineduc=83	0.660***	(0.051)
mineduc=84	0.208***	(0.050)
mineduc=85	0.872***	(0.046)
mineduc=86	1.548***	(0.063)
mineduc=87	1.169***	(0.050)
mineduc=88	1.519***	(0.057)
mineduc=89	1.047***	(0.052)
mineduc=90	0.604***	(0.047)
mineduc=91	1.134***	(0.057)
mineduc=92	0.560***	(0.049)
mineduc=93	0.356***	(0.048)
mineduc=94	0.383***	(0.044)
mineduc=990	0.801***	(0.044)
mineduc=992	0.666***	(0.050)
mineduc=994	0.317***	(0.023)
mineduc=995	0.253***	(0.013)
Observations	1,098,085	
Mean of Dep. Variable	0.218	

Standard errors in parentheses

Logistic regression of choosing the outside option (dropping out) in period t. Controls include vector of characteristics, Institution and major FE, year enrolled FE and cumulated performance. PSU and GPU are standardised. Base category is low SES and t=1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B3: Performance probabilities

	index	
female	0.442***	(0.019)
Middle SES	0.063***	(0.024)
High SES	0.069***	(0.023)
PSU	0.949***	(0.022)
GPA	0.331***	(0.012)
OP fees	-0.011***	(0.002)
Rel. Ability	-0.204***	(0.009)
Distance	-0.005	(0.010)
Cum. credits completed (t-1)	-0.005**	(0.002)
Credits completed (t-1)	0.523***	(0.005)
1 cred. left (t-1)	-6.193***	(0.121)
2 cred. left (t-1)	-2.462***	(0.039)
3 cred. left (t-1)	-1.950***	(0.043)
Delay 4+ (t-1)	-1.067***	(0.012)
OP × female	0.015***	(0.002)
OP × middle SES	0.024***	(0.003)
OP × high SES	-0.001	(0.003)
OP × psu	0.006***	(0.001)
OP × gpa	0.000	(0.001)
1 left (t-1) × female	-0.053	(0.132)
1 left (t-1) × middle SES	-0.122	(0.170)
1 left (t-1) × high SES	-0.150	(0.162)
1 left (t-1) × psu	-0.603***	(0.084)
1 left (t-1) × gpa	-0.245***	(0.083)
2 left (t-1) × female	-0.038	(0.042)
2 left (t-1) × middle SES	-0.128**	(0.056)
2 left (t-1) × high SES	0.045	(0.049)
2 left (t-1) × psu	-0.176***	(0.028)
2 left (t-1) × gpa	0.021	(0.026)
3 left (t-1) × female	-0.018	(0.047)
3 left (t-1) × middle SES	-0.010	(0.063)
3 left (t-1) × high SES	0.144***	(0.055)
3 left (t-1) × psu	-0.132***	(0.033)
3 left (t-1) × gpa	-0.020	(0.029)
Delay (t-1) × female	-0.065***	(0.011)
Delay (t-1) × middle SES	0.043***	(0.014)
Delay (t-1) × high SES	0.089***	(0.013)
Delay (t-1) × psu	-0.058***	(0.007)
Delay (t-1) × gpa	0.043***	(0.007)
t=2 × SCP	-0.340***	(0.015)
t=3 × SCP	-1.493***	(0.016)
t=4 × SCP	-1.563***	(0.020)
t=5 × SCP	-2.126***	(0.028)
t=6 × SCP	-1.543***	(0.044)
t=7 × SCP	-0.912***	(0.087)
Credits (t-1) × female	-0.010***	(0.003)
Credits (t-1) × middle SES	-0.013***	(0.004)
Credits (t-1) × high SES	0.004	(0.004)
Credits (t-1) × psu	-0.058***	(0.003)
Credits (t-1) × gpa	-0.005**	(0.002)
Cumul (t-1) × female	-0.032***	(0.001)
Cumul (t-1) × middle SES	-0.006***	(0.002)
Cumul (t-1) × high SES	-0.012***	(0.001)
Cumul (t-1) × psu	-0.011***	(0.001)
Cumul (t-1) × gpa	-0.014***	(0.001)
t= 2	-1.107***	(0.016)
t= 3	-0.416***	(0.019)
t= 4	0.022	(0.024)
t= 5	-0.007	(0.030)
t= 6	-0.471***	(0.036)
t= 7	-0.660***	(0.041)
Observations	1,097,591	
Mean of Dep. Variable	3.097	

Standard errors in parentheses

Ordered logistic regression of discretized credit achievement (0-4). Controls include vector of characteristics, Institution and major FE, year enrolled FE and cumulated performance. PSU and GPU are standardized. Base category is low SES and t=1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B4: Performance Probabilities (extended)

	index	
female	0.442***	(0.019)
Middle SES	0.063***	(0.024)
High SES	0.069***	(0.023)
PSU	0.949***	(0.022)
GPA	0.331***	(0.012)
OP fees	-0.011***	(0.002)
Rel. Ability	-0.204***	(0.009)
Distance	-0.005	(0.010)
Cum. credits completed (t-1)	-0.005**	(0.002)
Credits completed (t-1)	0.523***	(0.005)
1 cred. left (t-1)	-6.193***	(0.121)
2 cred. left (t-1)	-2.462***	(0.039)
3 cred. left (t-1)	-1.950***	(0.043)
Delay 4+ (t-1)	-1.067***	(0.012)
OP × female	0.015***	(0.002)
OP × middle SES	0.024***	(0.003)
OP × high SES	-0.001	(0.003)
OP × psu	0.006***	(0.001)
OP × gpa	0.000	(0.001)
1 left (t-1) × female	-0.053	(0.132)
1 left (t-1) × middle SES	-0.122	(0.170)
1 left (t-1) × high SES	-0.150	(0.162)
1 left (t-1) × psu	-0.603***	(0.084)
1 left (t-1) × gpa	-0.245***	(0.083)
2 left (t-1) × female	-0.038	(0.042)
2 left (t-1) × middle SES	-0.128**	(0.056)
2 left (t-1) × high SES	0.045	(0.049)
2 left (t-1) × psu	-0.176***	(0.028)
2 left (t-1) × gpa	0.021	(0.026)
3 left (t-1) × female	-0.018	(0.047)
3 left (t-1) × middle SES	-0.010	(0.063)
3 left (t-1) × high SES	0.144***	(0.055)
3 left (t-1) × psu	-0.132***	(0.033)
3 left (t-1) × gpa	-0.020	(0.029)
Delay (t-1) × female	-0.065***	(0.011)
Delay (t-1) × middle SES	0.043***	(0.014)
Delay (t-1) × high SES	0.089***	(0.013)
Delay (t-1) × psu	-0.058***	(0.007)
Delay (t-1) × gpa	0.043***	(0.007)
t=2 × SCP	-0.340***	(0.015)
t=3 × SCP	-1.493***	(0.016)
t=4 × SCP	-1.563***	(0.020)
t=5 × SCP	-2.126***	(0.028)
t=6 × SCP	-1.543***	(0.044)
t=7 × SCP	-0.912***	(0.087)

Credits (t-1) × female	-0.010***	(0.003)
Credits (t-1) × middle SES	-0.013***	(0.004)
Credits (t-1) × high SES	0.004	(0.004)
Credits (t-1) × psu	-0.058***	(0.003)
Credits (t-1) × gpa	-0.005**	(0.002)
Cumul (t-1) × female	-0.032***	(0.001)
Cumul (t-1) × middle SES	-0.006***	(0.002)
Cumul (t-1) × high SES	-0.012***	(0.001)
Cumul (t-1) × psu	-0.011***	(0.001)
Cumul (t-1) × gpa	-0.014***	(0.001)
Rel. Abil. x female	0.010**	(0.005)
Rel. Abil. x middle SES	0.008	(0.006)
Rel. Abil. x high SES	-0.038***	(0.006)
Rel. Abil. x psu	-0.030***	(0.003)
Rel. Abil. x gpa	0.034***	(0.003)
Dist. x female	0.006	(0.011)
Dist. x middle SES	-0.049***	(0.015)
Dist. x high SES	-0.073***	(0.014)
Dist. x psu	-0.040***	(0.008)
Dist. x gpa	0.009	(0.007)
Agricultural × fem	-0.310***	(0.033)
Agricultural × middle SES	-0.099**	(0.043)
Agricultural × high SES	0.054	(0.038)
Agricultural × psu	0.016	(0.027)
Agricultural × gpa	-0.025	(0.021)
Arts and Architecture × fem	-0.090***	(0.029)
Arts and Architecture × middle SES	0.023	(0.039)
Arts and Architecture × high SES	0.263***	(0.034)
Arts and Architecture × psu	-0.120***	(0.024)
Arts and Architecture × gpa	0.002	(0.018)
Science × fem	-0.208***	(0.030)
Science × middle SES	-0.055	(0.040)
Science × high SES	0.150***	(0.035)
Science × psu	-0.107***	(0.024)
Science × gpa	-0.155***	(0.019)
Social Sciences × fem	0.095***	(0.024)
Social Sciences × middle SES	-0.017	(0.031)
Social Sciences × high SES	-0.018	(0.028)
Social Sciences × psu	-0.114***	(0.019)
Social Sciences × gpa	0.024	(0.015)
Law × fem	-0.170***	(0.026)
Law × middle SES	-0.015	(0.035)
Law × high SES	0.158***	(0.030)
Law × psu	0.023	(0.021)
Law × gpa	-0.078***	(0.016)
Education × fem	0.170***	(0.025)
Education × middle SES	0.062**	(0.031)

Education $\times$ high SES	0.196***	(0.030)
Education $\times$ psu	-0.174***	(0.020)
Education $\times$ gpa	-0.007	(0.015)
Humanities $\times$ fem	-0.133**	(0.067)
Humanities $\times$ middle SES	0.012	(0.077)
Humanities $\times$ high SES	0.084	(0.075)
Humanities $\times$ psu	0.152***	(0.052)
Humanities $\times$ gpa	-0.014	(0.041)
SCP $\times$ fem	0.070*	(0.042)
SCP $\times$ middle SES	0.012	(0.050)
SCP $\times$ high SES	0.157***	(0.056)
SCP $\times$ psu	-0.243***	(0.029)
SCP $\times$ gpa	-0.040	(0.025)
SCP-Gratuidad $\times$ fem	-0.002	(0.019)
SCP-Gratuidad $\times$ middle SES	0.039	(0.025)
SCP-Gratuidad $\times$ high SES	0.144***	(0.024)
SCP-Gratuidad $\times$ psu	-0.147***	(0.016)
SCP-Gratuidad $\times$ gpa	0.081***	(0.012)
Health $\times$ fem	-0.051**	(0.021)
Health $\times$ middle SES	-0.030	(0.027)
Health $\times$ high SES	0.106***	(0.024)
Health $\times$ psu	-0.014	(0.016)
Health $\times$ gpa	-0.002	(0.012)
Technology $\times$ fem	-0.105***	(0.021)
Technology $\times$ middle SES	-0.021	(0.027)
Technology $\times$ high SES	0.154***	(0.024)
Technology $\times$ psu	0.079***	(0.016)
Technology $\times$ gpa	-0.020*	(0.012)
t= 2	-1.107***	(0.016)
t= 3	-0.416***	(0.019)
t= 4	0.022	(0.024)
t= 5	-0.007	(0.030)
t= 6	-0.471***	(0.036)
t= 7	-0.660***	(0.041)
Agricultural	-0.216***	(0.032)
Arts and Architecture	-0.152***	(0.029)
Science	-0.467***	(0.029)
Social Sciences	0.267***	(0.023)
Law	-0.424***	(0.026)
Education	0.176***	(0.023)
Humanities	-0.329***	(0.066)
SCP	0.051	(0.057)
SCP-Gratuidad	0.572***	(0.029)
Health	-0.241***	(0.021)
Technology	-0.574***	(0.019)
mineduc=2	-0.860***	(0.033)
mineduc=3	-1.275***	(0.035)

mineduc=4	-0.357***	(0.028)
mineduc=9	-0.157***	(0.038)
mineduc=10	-0.612***	(0.029)
mineduc=11	-0.841***	(0.060)
mineduc=13	-1.855***	(0.032)
mineduc=20	-1.757***	(0.029)
mineduc=22	-0.626***	(0.028)
mineduc=23	-1.369***	(0.041)
mineduc=31	-0.306***	(0.026)
mineduc=34	-1.642***	(0.042)
mineduc=39	-0.492***	(0.025)
mineduc=42	-0.456***	(0.035)
mineduc=45	-1.063***	(0.031)
mineduc=50	0.016	(0.032)
mineduc=54	0.203***	(0.028)
mineduc=69	-1.168***	(0.036)
mineduc=70	-1.502***	(0.039)
mineduc=71	-1.288***	(0.033)
mineduc=72	-0.918***	(0.030)
mineduc=73	-0.783***	(0.031)
mineduc=74	-1.312***	(0.033)
mineduc=75	-0.800***	(0.032)
mineduc=76	-1.026***	(0.032)
mineduc=77	-0.961***	(0.039)
mineduc=78	-1.138***	(0.033)
mineduc=79	-1.082***	(0.033)
mineduc=80	-0.818***	(0.030)
mineduc=81	-0.806***	(0.035)
mineduc=82	-1.361***	(0.042)
mineduc=83	-0.993***	(0.033)
mineduc=84	-1.019***	(0.032)
mineduc=85	-0.721***	(0.030)
mineduc=86	-1.706***	(0.041)
mineduc=87	-1.254***	(0.032)
mineduc=88	-1.665***	(0.037)
mineduc=89	-1.357***	(0.033)
mineduc=90	-1.226***	(0.031)
mineduc=91	-1.596***	(0.037)
mineduc=92	-0.793***	(0.031)
mineduc=93	-0.994***	(0.031)
mineduc=94	-0.861***	(0.028)
mineduc=990	-0.323***	(0.030)
mineduc=992	0.129***	(0.037)
mineduc=994	-0.498***	(0.016)
mineduc=995	-0.108***	(0.009)
cut1	-2.716***	(0.030)
cut2	-2.164***	(0.030)

cut3	-1.264***	(0.030)
cut4	-0.878***	(0.030)
Observations	1,097,591	
Mean of Dep. Variable	3.097	

Standard errors in parentheses

Ordered logistic regression of discretized credit achievement (0-4).Ordered logistic regression of discretized credit achievement (0-4). Controls include vector of characteristics, Institution and major FE, year enrolled FE and cumulated performance. PSU and GPU are standardized. Base category is low SES and t=1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

C Estimation

C.1 Functional forms

C.1.1 Performance outcomes

For $g \in \mathcal{G} = \{0, 1, 2, 3, 4\}$, denote $\tilde{g}$ the underlying index:

$$\begin{aligned} \tilde{g}_j(x_{it}) = & \sum_m \sum_n \alpha^g_{mn} Z_{im} Z_{jn} + \sum_m \sum_n \alpha^g_{gm} Z_{gm} Z_{in} + \mathbf{Z_g}' \alpha^g_G \\ & + \alpha^g_{field(j)} + \alpha^g_{inst(j)} + \lambda^g_t + \lambda^g_{tscp} \cdot \mathrm{SCP}_{ij}, \end{aligned} \tag{13}$$

where Z_i = (psu, gpa, income, female), Z_j = (field1, ..., field10), Z_g = (g_{it-1}, G_{it-1}, delay$_{it-1}$, 1left$_{it-1}$, 2left$_{it-1}$, 3left$_{it-1}$). g_{it-1} and G_{it-1} denote previous-period and cumulative credit completion, respectively. Xleft indicates whether the student left the program with X or more credits remaining. $\alpha^g_{field(j)}$ are field fixed effects, $\alpha^g_{inst(j)}$ are institution fixed effects, and λ^g_t are period fixed effects. Period fixed effects are allowed to differ between SCPs and university programs.

Xleft variables take into account the fact that passing all the credits in a period where less credits are needed to graduate might be different. An alternative modelling approach would estimate separate models for each possible credit left in a year (1,2,3,4). However, estimating it together gives us more precise estimates.

C.1.2 Dropout probability

We estimate the following logistic regression by maximum likelihood:

$$\begin{aligned} \tilde{drop}_j(x_{it}) = & \sum_m \sum_n \alpha^d_{mn} Z_{im} Z_{jn} + \sum_m \sum_n \alpha^d_{gm} Z_{gm} Z_{in} + \mathbf{Z_g}' \alpha^d_G \\ & + \alpha^d_{field(j)} + \alpha^d_{inst(j)} + \lambda^d_t + \lambda^d_{tscp} \cdot \mathrm{SCP}_{ij}, \end{aligned} \tag{14}$$

where Z_i = (psu, gpa, income, female), Z_j = (field1, ..., field10), Z_g = (g_{it-1}, G_{it-1}, delay$_{it-1}$, 1left$_{it-1}$, 2left$_{it-1}$, 3left$_{it-1}$). g_{it-1} and G_{it-1} denote previous-period and cumulative credit completion, respectively. Xleft indicates whether the student left the program with X or more credits remaining. $\alpha^g_{field(j)}$ are field fixed effects, $\alpha^g_{inst(j)}$ are institution fixed effects, and λ^g_t are period fixed effects. Period fixed effects are allowed to differ between SCPs and university programs.

C.1.3 Labour market

We estimate the following wage equation,

$$\log(w_j(x_{it})) = \alpha_0^w + \alpha_f^w \text{ female}_i + \alpha_e^w \text{ exper}_{it} + \alpha_{e2}^w \text{ exper}_{it}^2 + \alpha_j^w + \lambda_R^w + \sum_{r=1}^{16} \sum_j \alpha_{rj}^w R_{ir} \cdot 1\{field(j)\} + \varepsilon_{ijt}, \tag{15}$$

where R_{ir} is a dummy for region r. The region–field interactions $\{R_{ir} \cdot 1\{field(j)\}\}$ are crucial for identification of the wage coefficient in the structural model, as they serve as an exclusion restriction: regional-field variation affects utility only through wages.

From scraped data on wages from the [6]Chilean Ministry of Education, we recover institution-specific wage premiums by regressing log wages on field and institution fixed effects. We then predict Equation 15 for each student in the sample, adding the institution premium.

C.1.4 Fixed costs

$$FC_j(x_{it}) = \alpha_p OP_{ij} + \sum_m \alpha_{pm} OP_{ij} Z_{im} + \sum_m \sum_n \alpha_{mn} Z_{im} Z_{jn} + \sum_m \sum_n \alpha_{gm} Z_{gm} Z_{in} + \mathbf{Z_g}' \alpha_G + \sum_m \sum_k \alpha_{km} M_{ij,k} Z_{im} + \mathbf{M_{ij}}' \alpha_m + \alpha_{field(j)} + \alpha_{inst(j)}, \tag{16}$$

where D_{ij} is a dummy for the institution being in a different region (distance proxy), and A_{ij} is the student's relative ability in program j. $M_{ij} = (D_{ij}, A_{ij})$ is a vector of student–program matched variables.

[6]https://www.mifuturo.cl

C.2 Conditional Value Function

Conditional value function making use of a terminal action (dropping out) when log effort is the underlying index of an ordered logit model.

$$
\begin{aligned}
v_j(x_{it}, e_{it}) &= u_j(x_{it}, e_{it}) + \beta \sum_{\bar{g} \in \mathcal{G}} \phi^{\bar{g}}(e_{it}) \left(\gamma + v_0(x_{it+1}) - \ln \Pr(d^0_{it+1} | x_{it+1}(\bar{g})\right) \\
v_j(x_{it}, e_{it}) &= -FC(x_{it}, OP) - c(x_{it}) e(x_{it}) + \beta \sum_{\bar{g} \in \mathcal{G}} \phi^{\bar{g}}(e_{it}) \left(\gamma + v_0(x_{it+1}) - \ln \Pr(d^0_{it+1} | x_{it+1}(\bar{g}))\right) \\
v_j(x_{it}, e_{it}) &= -FC(x_{it}, OP) - \beta \sum_{\bar{g} \in \mathcal{G}} \frac{\partial \phi^{\bar{g}_{ijt}}(e_{it})}{\partial e_{it}} \left(\gamma + v_0(x_{it+1}) - \ln \Pr(d^0_{it+1} | x_{it+1}(\bar{g}))\right) e(x_{it}) \\
&\quad + \beta \sum_{\bar{g} \in \mathcal{G}} \phi^{\bar{g}}(e_{it}) \left(\gamma + v_0(x_{it+1}) - \ln \Pr(d^0_{it+1} | x_{it+1}(\bar{g}))\right) \\
v_j(x_{it}, e_{it}) &= -FC(x_{it}, OP) + \beta\gamma \\
&\quad + \beta \sum_{\bar{g} \in \mathcal{G}} \left(\left(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1} | x_{it+1}(\bar{g}))\right) \times \left(\phi^{\bar{g}}(e^*_j(x_{it})) - \frac{\partial \phi^{\bar{g}_{ijt}}(e_{it})}{\partial e_{it}} \Big|_{e_{it} = e^*_j(x_{it})} e^*_j(x_{it}) \right) \right) \\
v_j(x_{it}, e_{it}) &= -FC(x_{it}, OP) + \beta\gamma \\
&\quad + \beta \sum_{\bar{g} \in \mathcal{G}} \left(\left(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1} | x_{it+1}(\bar{g}))\right) \right. \\
&\quad \times \left(\Lambda(\alpha_{\bar{g}} - e^*_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - e^*_j(x_{it})\right) \\
&\quad - \left(\Lambda(\alpha_{\bar{g}-1} - e^*_j(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}-1} - e^*_j(x_{it}))) - \Lambda(\alpha_{\bar{g}} - e^*_j(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}} - e^*_j(x_{it})))\right) \\
&\quad \left. \times e^*_j(x_{it}))\right)
\end{aligned}
$$

where $\Lambda(\cdot)$ denotes the standard logistic CDF. The general expression for ordered logit probabilities are, denoting by $\alpha_{\bar{g}}$ the estimated cutoff points,

$$
\begin{aligned}
\phi^{\bar{g}}_{ijt}(e^*_j(x_{it})) &= \Lambda(\alpha_{\bar{g}} - e^*_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - e^*_j(x_{it})) \\
&= \frac{1}{1 + \exp(e^*_j(x_{it}) - \alpha_{\bar{g}})} - \frac{1}{1 + \exp(e^*_j(x_{it}) - \alpha_{\bar{g}-1})}
\end{aligned}
$$

The derivative wrt $e_j^*(x_{it})$ writes

$$\frac{\partial \phi_{ijt}^{\bar{g}}(e_{it})}{\partial e_{it}}\Big|_{e_{it}=e_j^*(x_{it})} = -\Lambda(\alpha_{\bar{g}} - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}} - e_j^*(x_{it})))$$
$$- \left(-\Lambda(\alpha_{\bar{g}-1} - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}-1} - e_j^*(x_{it})))\right)$$
$$\frac{\partial \phi_{ijt}^{\bar{g}}(e_{it})}{\partial e_{it}}\Big|_{e_{it}=e_j^*(x_{it})} = \Lambda(\alpha_{\bar{g}-1} - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}-1} - e_j^*(x_{it})))$$
$$- \Lambda(\alpha_{\bar{g}} - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}} - e_j^*(x_{it})))$$

Computing the marginal cost and EMAX expression

$$\sum_{\bar{g}\in\mathcal{G}} \left(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d_{it+1}^0 | x_{it+1}(\bar{g}))\right) \times \left(\phi_{ijt}^{\bar{g}}(e_j^*(x_{it})) - \frac{\partial \phi_{ijt}^{\bar{g}}(e_{it})}{\partial e_{it}}\Big|_{e_{it}=e_j^*(x_{it})} e_j^*(x_{it})\right)$$

For $g = 1$

$$\left(\Lambda(\alpha_1 - e_j^*(x_{it})) + \left(\Lambda(\alpha_1 - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_1 - e_j^*(x_{it})))\right) \times e_j^*(x_{it})\right)$$
$$\left(\Lambda(\alpha_1 - e_j^*(x_{it})) + \left(\Lambda(\alpha_1 - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_1 - e_j^*(x_{it})))\right) \times e_j^*(x_{it})\right)$$

$$\left(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d_{it+1}^0 | x_{it+1}(1))\right) \times$$
$$\left(\Lambda(\alpha_1 - index_j(x_{it})) - \frac{\partial \phi_{ijt}^{\bar{g}}(e_j(x_{it}))}{\partial e_{it}}\Big|_{e_{it}=e_j^*(x_{it})} \times e_j^*(x_{it})\right)$$

We defined effort as

$$e_j(x_{it}) = \frac{1 - \Lambda(\alpha_1 - index_j(x_{it}))}{\Lambda(\alpha_1 - index_j(x_{it}))} \tag{17}$$
$$\Lambda(\alpha_1 - index_j(x_{it})) = \frac{1}{1 + e_j(x_{it})} \tag{18}$$
$$\alpha_1 - index_j(x_{it}) = \ln\left(\frac{\frac{1}{1+e_j(x_{it})}}{1 - \frac{1}{1+e_j(x_{it})}}\right) \tag{19}$$
$$\alpha_1 - index_j(x_{it}) = \ln\left(\frac{1}{e_j(x_{it})}\right) = -\ln(e_j(x_{it})) \tag{20}$$
$$\alpha_1 - \ln(e_j(x_{it})) = index_j(x_{it}) \tag{21}$$

then the derivative wrt effort looks like

$$\begin{aligned}
&\left(\Lambda(\alpha_1 - index_j(x_{it})) - \left(\lambda(\alpha_1 - index_j(x_{it})) \times \frac{-1}{e_j(x_{it})}\right) \times e_j(x_{it})\right) \\
&= \frac{1}{1+e_j(x_{it})} + \frac{e_j(x_{it})}{(1+e_j(x_{it}))^2}
\end{aligned}$$

For $g \in (2, 4)$

$$\begin{aligned}
&\left(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1}|x_{it+1}(\bar{g}))\right) \times \\
&(\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) - \\
&(\Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it}))) - \Lambda(\alpha_{\bar{g}} - index_j(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}} - index_j(x_{it})))) \times
\end{aligned}$$

Recall that rearranging the effort measure in Equation 17, we can express $index_j(x_{it}) = \alpha_1 - \ln(e_j(x_{it}))$

$$\begin{aligned}
&\left(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1}|x_{it+1}(\bar{g}))\right) \times \\
&(\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) - \\
&\left.\frac{\partial \phi^{\bar{g}}_{ijt}(e_j(x_{it}))}{\partial e_{it}}\right|_{e_{it}=e^*_j(x_{it})} (\Lambda(\alpha_{\bar{g}} - \alpha_1 - \ln(e^*_j(x_{it}))) - \Lambda(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e^*_j(x_{it}))) \times e^*_j(x_{it})\Big) \\
&= \left(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1}|x_{it+1}(\bar{g}))\right) \times \\
&(\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) - \\
&\left(\lambda(\alpha_{\bar{g}} - \alpha_1 - \ln(e^*_j(x_{it})) \times \left(\frac{-1}{e^*_j(x_{it})}\right) - \lambda(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e^*_j(x_{it}))) \times \left(\frac{-1}{e^*_j(x_{it})}\right)\right) \times e^*_j(x_{it})\Big) \\
&= \left(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1}|x_{it+1}(\bar{g}))\right) \times \\
&(\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) - \\
&\left(\frac{-e^*_j(x_{it})}{e^*_j(x_{it})}\right) \times \left(\frac{\exp(\alpha_{\bar{g}} - \alpha_1 - \ln(e^*_j(x_{it})))}{(1+\exp(\alpha_{\bar{g}} - \alpha_1 - \ln(e^*_j(x_{it}))))^2} - \frac{\exp(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e^*_j(x_{it})))}{(1+\exp(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e^*_j(x_{it}))))^2}\right)\Big) \\
&= \left(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1}|x_{it+1}(\bar{g}))\right) \times \\
&(\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) + \\
&\left(\frac{\exp(\alpha_{\bar{g}} - \alpha_1 - \ln(e^*_j(x_{it})))}{(1+\exp(\alpha_{\bar{g}} - \alpha_1 - \ln(e^*_j(x_{it}))))^2} - \frac{\exp(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e^*_j(x_{it})))}{(1+\exp(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e^*_j(x_{it}))))^2}\right)\Big)
\end{aligned}$$

For $g = 5$

$$\big(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1}|x_{it+1}(5))\big) \times$$
$$\big(1 - \Lambda(\alpha_4 - e^*_j(x_{it})) - \big(\Lambda(\alpha_4 - e^*_j(x_{it})) \times (1 - \Lambda(\alpha_4 - e^*_j(x_{it})))\big) \times e^*_j(x_{it})\big)$$

$$\sum_{\bar{g}\in\mathcal{G}} \left(\phi^{\bar{g}}_{ijt}(e^*_j(x_{it})) - \frac{\partial \phi^{\bar{g}}_{ijt}(e_{it})}{\partial e_{it}}\Big|_{e_{it}=e^*_j(x_{it})} e^*_j(x_{it})\right)$$

Notice from the above expression that

$$\sum_{\bar{g}\in\mathcal{G}} \big(\phi^{\bar{g}}_{ijt}(e^*_j(x_{it}))\big) = 1$$

Since we are integrating over all possible states and that

$$\sum_{\bar{g}\in\mathcal{G}} \left(\frac{\partial \phi^{\bar{g}}_{ijt}(e_{it})}{\partial e_{it}}\Big|_{e_{it}=e^*_j(x_{it})} e^*_j(x_{it})\right)$$
$$= \lambda(\alpha_1 - index(x_{it}) \times (-1/e^*_j(x_{it})) \times e^*_j(x_{it}) +$$
$$(\lambda(\alpha_2 - index(x_{it})) - \lambda(\alpha_1 - index(x_{it}))) \times (-1/e^*_j(x_{it})) \times e^*_j(x_{it}) +$$
$$(\lambda(\alpha_3 - index(x_{it}) - \lambda(\alpha_2 - index(x_{it}))) \times (-1/e^*_j(x_{it})) \times e^*_j(x_{it}) +$$
$$(\lambda(\alpha_4 - index(x_{it}) - \lambda(\alpha_3 - index(x_{it}))) \times (-1/e^*_j(x_{it})) \times e^*_j(x_{it}) +$$
$$(-\lambda(\alpha_4 - index(x_{it}))) \times (-1/e^*_j(x_{it})) \times e^*_j(x_{it})$$

since $index = \ln(e^*_j(x_{it})) + \alpha_1$ and that $\frac{\partial \lambda(\alpha_g - index(x_{it}))}{\partial e_{it}} = \lambda(\alpha_g - index(x_{it})) \times (-1/e^*_j(x_{it}))$.

This can be rewritten as

$$
\begin{aligned}
&\sum_{\bar{g}\in\mathcal{G}}\left(\frac{\partial\phi_{ijt}^{\bar{g}}(e_{it})}{\partial e_{it}}\Big|_{e_{it}=e_j^*(x_{it})}e_j^*(x_{it})\right)\\
&= -\lambda(\alpha_1 - index(x_{it})+\\
&- (\lambda(\alpha_2 - index(x_{it})) - \lambda(\alpha_1 - index(x_{it})))+\\
&- (\lambda(\alpha_3 - index(x_{it}) - \lambda(\alpha_2 - index(x_{it})))+\\
&- (\lambda(\alpha_4 - index(x_{it}) - \lambda(\alpha_3 - index(x_{it})))+\\
&- (-\lambda(\alpha_1 - index(x_{it})))\\
&= 0
\end{aligned}
$$

Conditional value function depending on t and j

$$
\begin{aligned}
v_j(x_{it}, e_{it}) = &-FC(x_{it}, OP) + \beta\gamma \\
&+ \beta \sum_{\bar{g}\in\mathcal{G}} \left(\left(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d^0_{it+1}|x_{it+1}(\bar{g}))\right) \times \left(\phi^{\bar{g}}(e^*_j(x_{it})) - \frac{\partial \phi^{\bar{g}_{ijt}}(e_{it})}{\partial e_{it}}\Big|_{e_{it}=e^*_j(x_{it})} e^*_j(x_{it})\right)\right)
\end{aligned}
$$

- if $t = 1$ & $j = 0$: $v_0(x_{it}, e_{it}) = \beta\alpha_w wage_0(x_{it})$
- it $t > 1$ & $j = 0$: $v_0(x_{it}, e_{it}) = \beta\alpha_w wage_0(x_{it})$
- if $t = last$ & $t \neq T$ & $d_{it} = d_{it-1}$: $v_j(x_{it}, e_{it}) = \beta\alpha_w wage_1(x_{it})$
- if $t = T$ & $d_{it} = d_{it-1}$: $v_j(x_{it}, e_{it}) = \beta\alpha_w(wage_0(x_{it})$ or $wage_1(x_{it}))$ depending on graduation (um accumulation variable).

In the last period ($T = 7$)

$$
\begin{aligned}
v_0(x_{it}, e_{it}) &= \alpha_w wage_0 \\
v_1(x_{it}, e_{it}) &= \alpha_w wage_1
\end{aligned}
$$